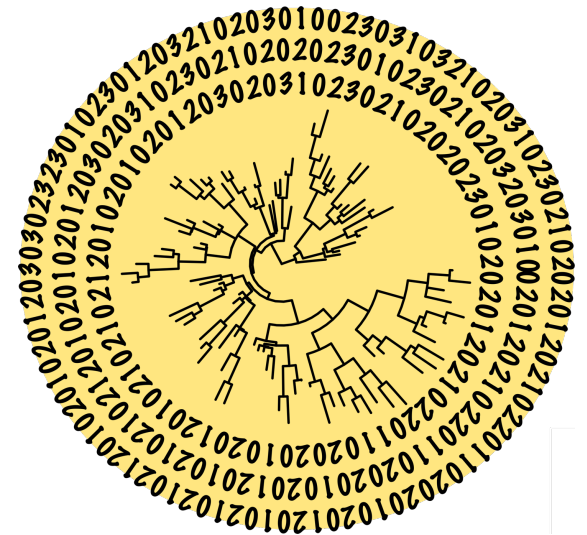


Morphological models and how to choose them

Laura Mulvey

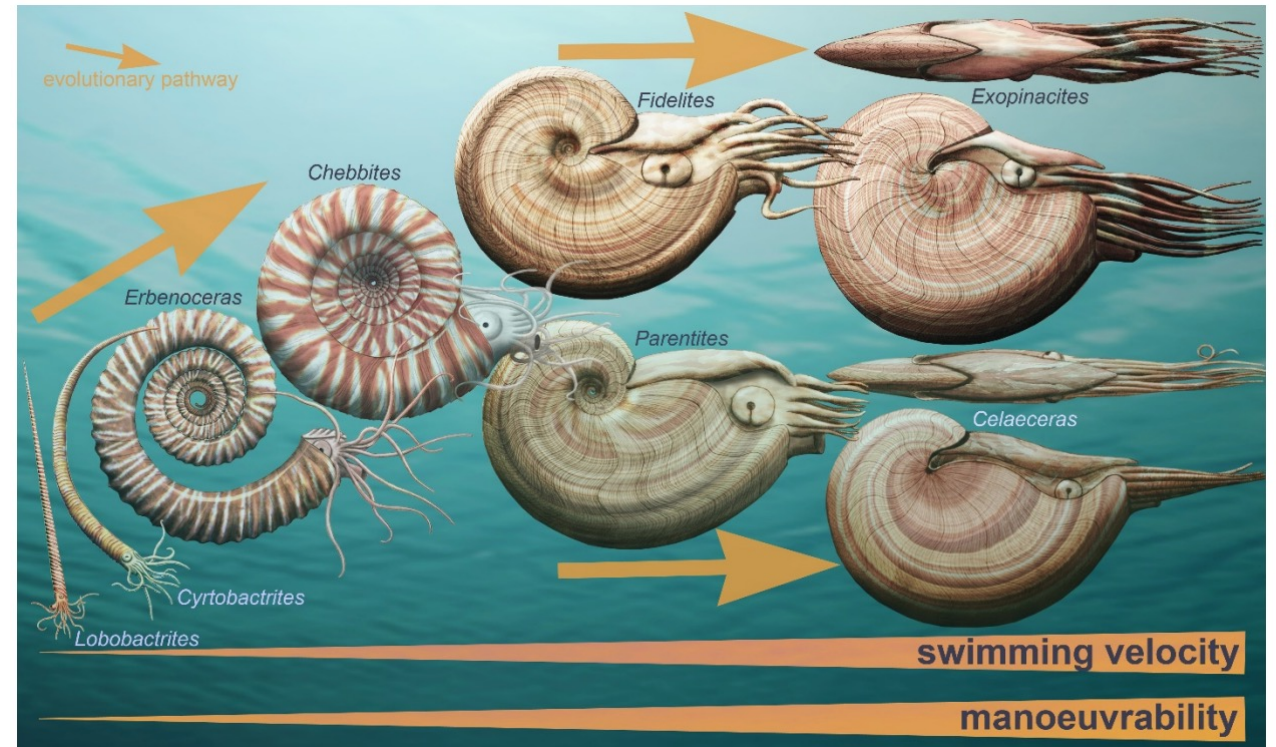
Phylogenetics Workshop



Morphological data

Morphological data was the original type of information used in phylogenetic analysis

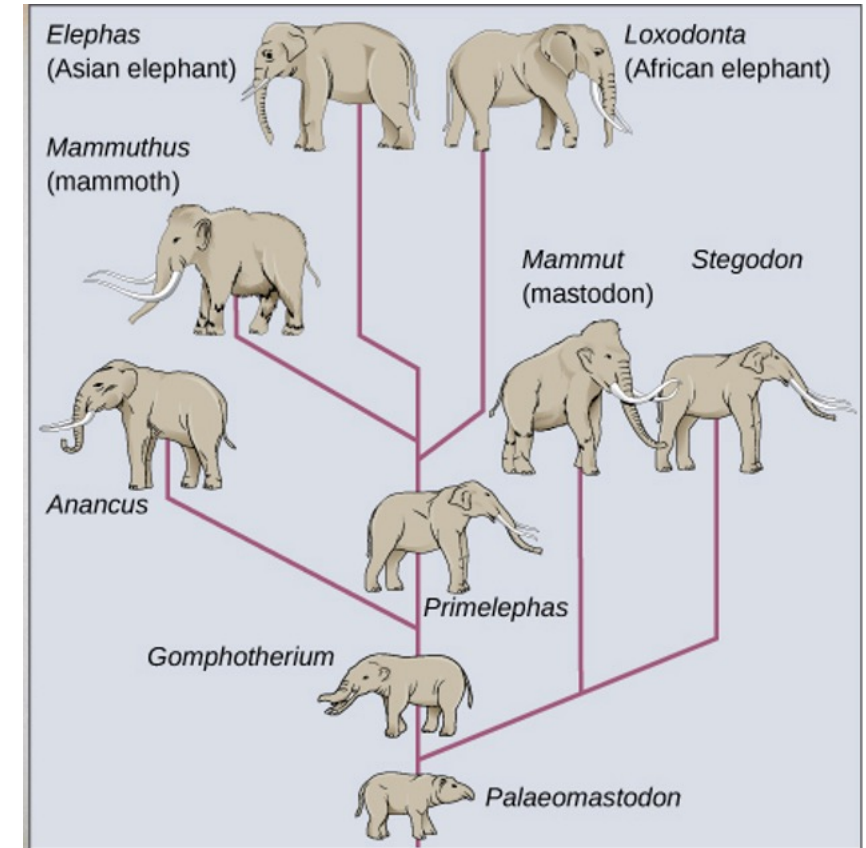
Fossils can be used to provide time calibrations, helps extant phylogeny, allows us to understand evolution through time



Morphological character data

Discrete Characters: Morphological data often consist of discrete characters, such as the presence or absence of certain traits, or more complex multistate traits (e.g., number of limbs, type of leaf, presence of a particular bone structure)

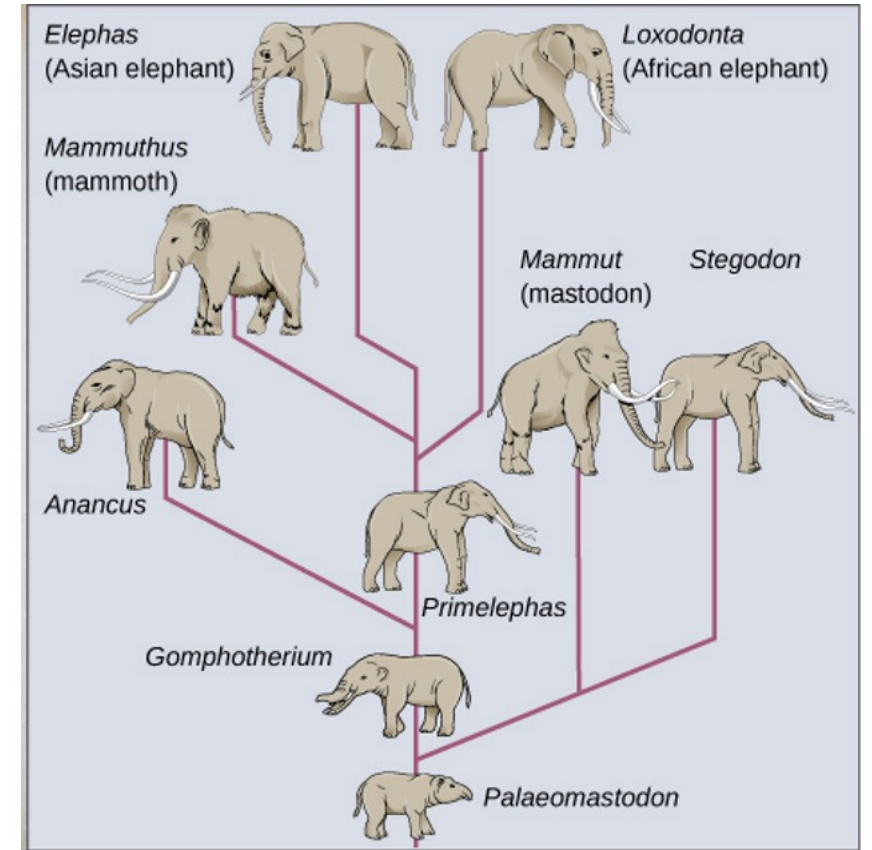
Continuous Characters: Some morphological data can be continuous, such as measurements of body size, length of bones, or other quantitative traits



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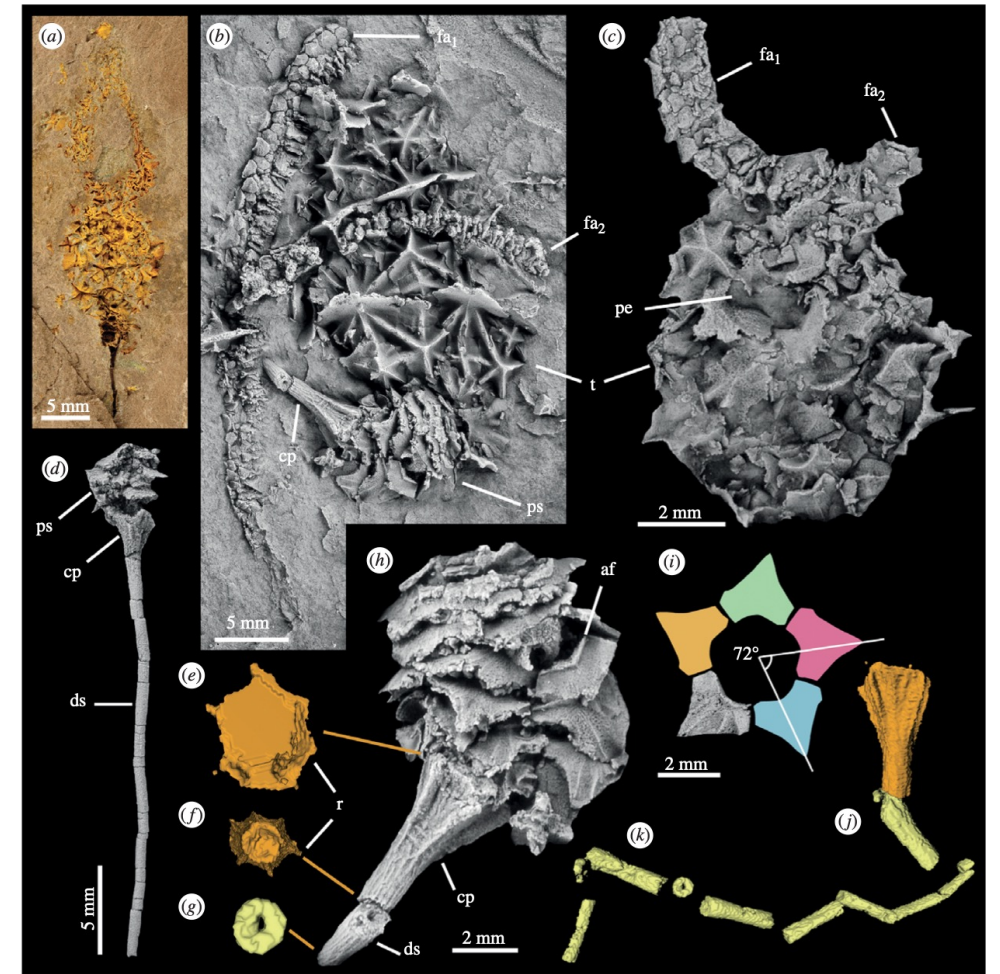
Continuous Characters: Some morphological data can be continuous, such as measurements of body size, length of bones, or other quantitative traits



Trait 1

Trait 28

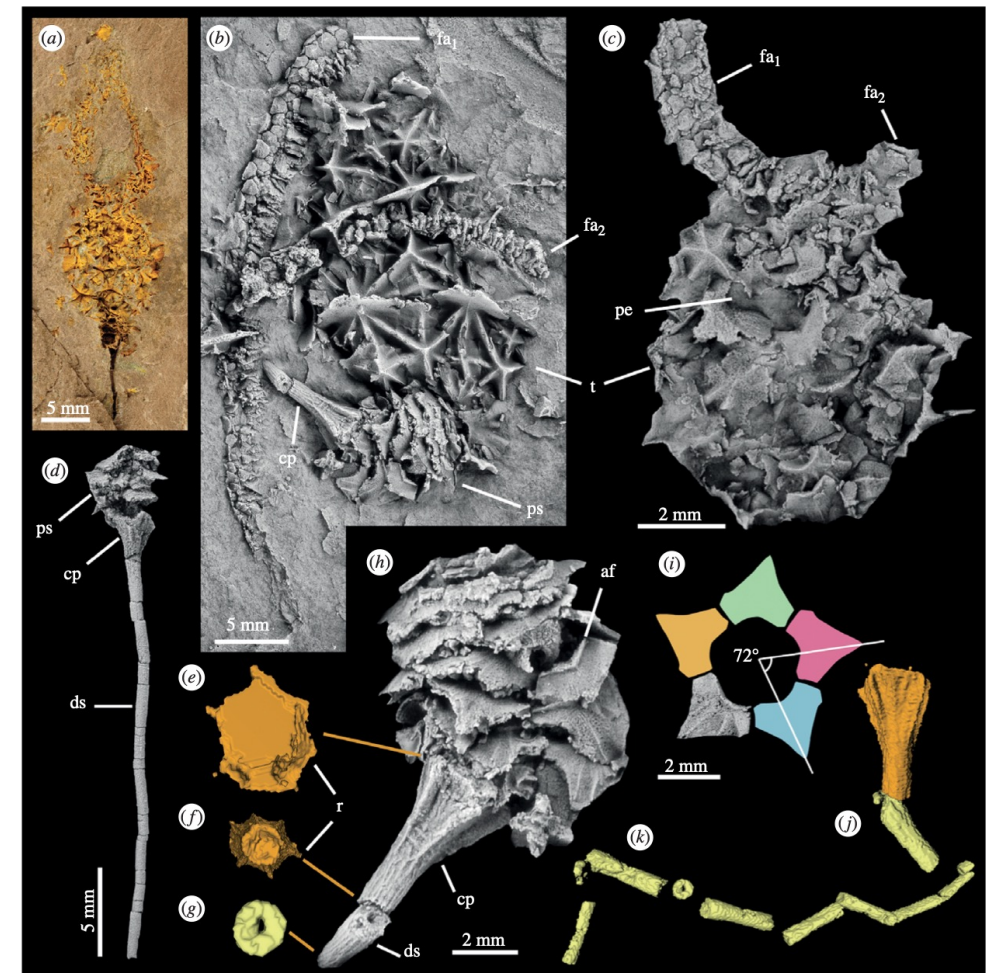
Taxa 1	001510010?00-100--00000000000
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	002500010?200100--0?100100000
	00?5?0010?200100?-0???010110
	0015000101201000430100011111
	0015000101201010440111011111
	??050?????201000440?11011111
	01050?010-210000?501??010110
	00020001002101003-1110010110
	0002000100211001441121011111
	000201111-210010?-??11011121
	?103?0?11?1001104-0000010000
	1005002110100010--0?00110?20
Taxa 14	1005002000101010540?00110020



Cambrian stalked echinoderms show unexpected plasticity of arm construction
Zamora & Smith. 2012 Proc B

Appendage branching pattern Cover plate arrangement Presence
Absence

001510010?00-100--00000000000
000500010?200100--00100100000
002500010?200100--0?100100000
00?5?0010?200100?-0???010110
0015000101201000430100011111
0015000101201010440111011111
??050? ??? ?201000440?11011111
01050?010-210000?501??010110
00020001002101003-1110010110
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?103?0?11?1001104-0000010000
1005002110100010--0?00110?20
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Cambrian stalked echinoderms show unexpected plasticity of arm construction
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Discrete character data

[illegible]

Discrete character data

Binary traits	0 1	Often describes the presence/absence of a trait
Multistate traits	0 1 2 3 4	Used to describe more complex traits and can capture greater variation between taxa

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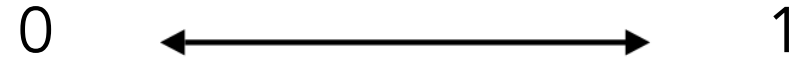
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Polymorphisms	0/1/2	Used when there are variations in a traits within species
Uncertain	0/1/2	Used when it is not clear which character trait is present in the taxon

How do we model
morphological
evolution?

Mk Model

Assumes **equal transition probabilities** between states

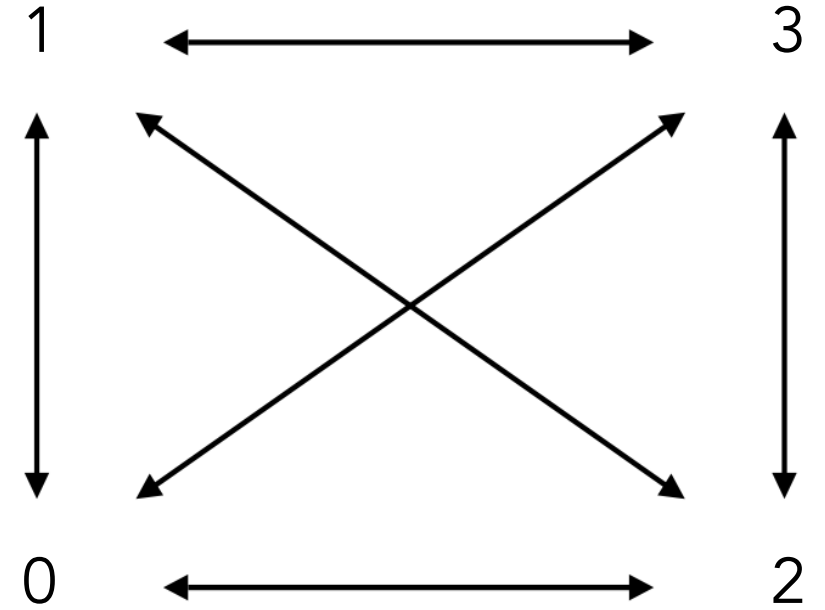


$$Q = \begin{bmatrix} -\mu_0 & \mu_{01} \\ \mu_{10} & -\mu_1 \end{bmatrix}$$

Mk Model

K can be any number of states

$$Q = \begin{pmatrix} -\mu_0 & \mu_{01} & \mu_{02} & \mu_{03} \\ \mu_{10} & -\mu_1 & \mu_{12} & \mu_{13} \\ \mu_{20} & \mu_{21} & -\mu_2 & \mu_{23} \\ \mu_{30} & \mu_{31} & \mu_{32} & -\mu_3 \end{pmatrix},$$



*4 state here as an example, can be any number from 2!

MkV model

What is one
characteristic of
morphological data
that is extremely
different to molecular

though there are
plenty.....

```
001510010?00-100--00000000000
000500010?200100--0010010000
002500010?200100--0?10010000
00?5?0010?200100?-0???010110
0015000101201000430100011111
0015000101201010440111011111
??050?????201000440?11011111
01050?010-210000?501??010110
00020001002101003-1110010110
0002000100211001441121011111
000201111-210010?-??11011121
?103?0?11?1001104-0000010000
1005002110100010--0?00110?20
1005002000101010540?00110020
```

MkV model

What is one
characteristic of
morphological data
that is extremely
different to molecular

though there are
plenty.....

All varying characters

```
001510010?00-100--00000000000
000500010?200100--00100100000
002500010?200100--0?100100000
00?5?0010?200100?-0???010110
0015000101201000430100011111
0015000101201010440111011111
??050?????201000440?11011111
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00020001002101003-1110010110
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?103?0?11?1001104-0000010000
1005002110100010--0?00110?20
1005002000101010540?00110020
```

MkV model



Corrects for **ascertainment bias**

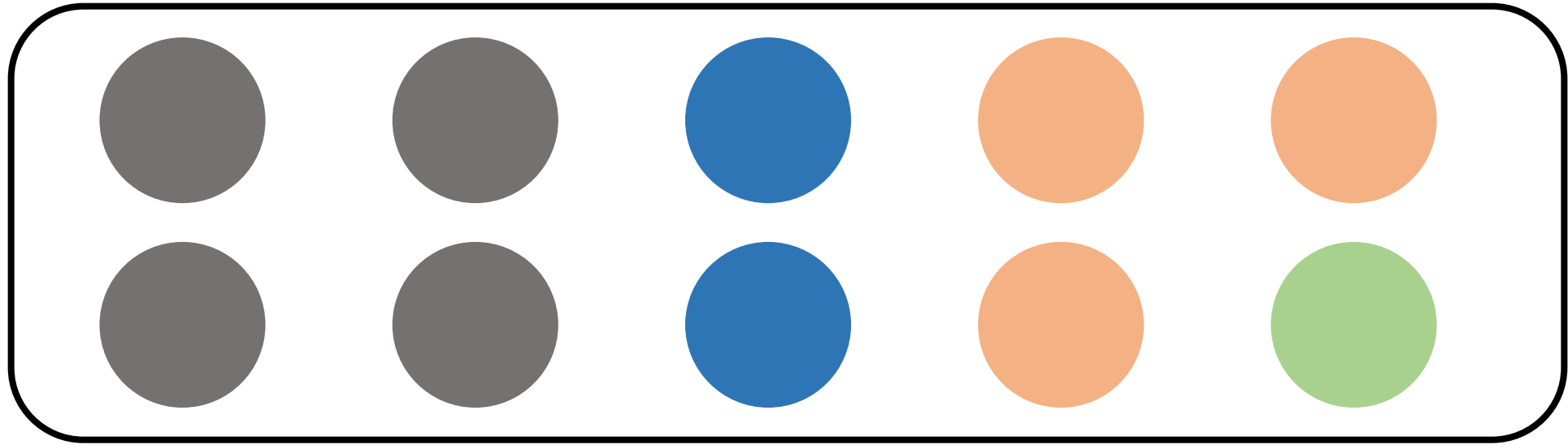
Failing to account for this can lead to **overestimations** in **branch lengths** and which can further lead to errors in topology!

Condition the likelihood
on there only being
varying site

$$\Pr (D \mid V) = \frac{\Pr (D,V)}{\Pr (V)}$$

MkV model

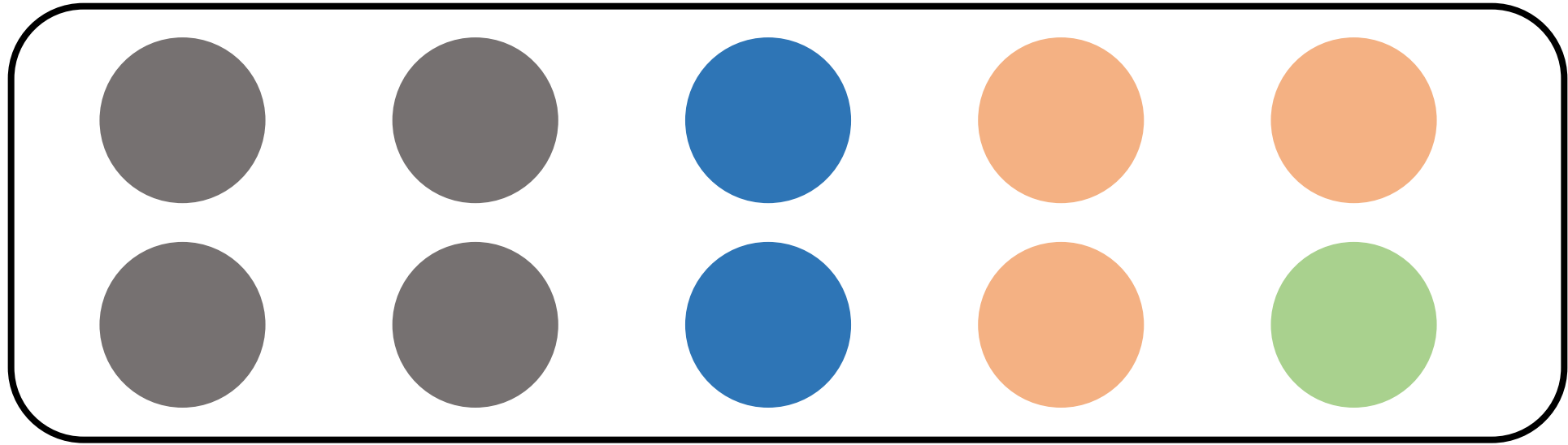
What is the probability for picking a certain colour ball?



$$1 = 0.4 + 0.2 + 0.3 + 0.1$$

MkV model

What is the probability for picking a certain colour ball?

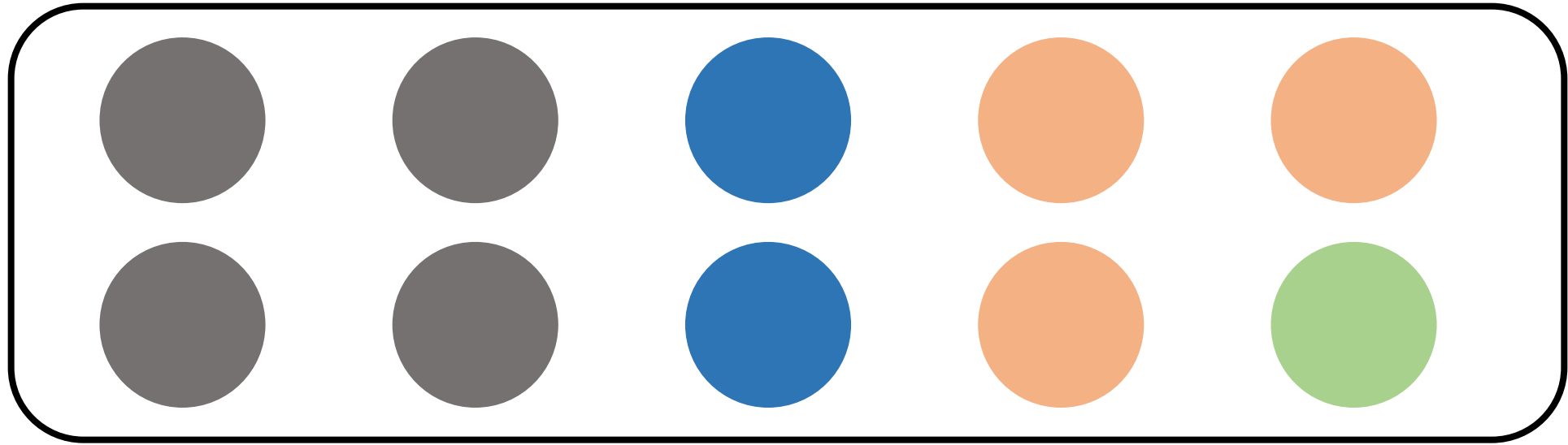


$$1 = 0.4 + 0.2 + 0.3 + 0.1$$

Probability of choosing an orange ball = 0.3

MkV model

What is the probability for picking a certain colour ball?



$$1 = 0.4 + 0.2 + 0.3 + 0.1$$

Probability of choosing an orange ball = 0.3

Probability of choosing an orange ball given it is not grey = $0.3/0.6 = 0.5$

*Adapted from Paul
Lewis PhyloSeminar*

MkV model

Corrects for **ascertainment bias**

Failing to account for this can lead to overestimations in branch lengths and which can further lead to errors in topology!



Probability of the data
given character is
variable



$$\Pr (D \mid V) = \frac{\Pr (D,V)}{\Pr (V)}$$

MkV model

Corrects for **ascertainment bias**

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Probability of the data
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variable



$$\Pr (D \mid V) = \frac{\Pr (D,V)}{\Pr (V)}$$

Probability of the
data and character
is variable



MkV model

Corrects for **ascertainment bias**



Failing to account for this can lead to overestimations in branch lengths and which can further lead to errors in topology!

Probability of the data
given character is
variable



$$\Pr(D | V) = \frac{\Pr(D, V)}{\Pr(V)}$$

Probability of the
data and character
is variable



$\Pr(V)$



Probability that
character is variable

MkV model

Corrects for **ascertainment bias**

Failing to account for this can lead to overestimations in branch lengths and which can further lead to errors in topology!



$$\Pr(D | V) = \frac{\Pr(D, V)}{\Pr(V)}$$

$\Pr(V)$

Probability that
character is variable

$1 - \Pr(\text{character is constant})$

This value, $\Pr(C)$ can be obtained using a **dummy character** having the same state for all internal nodes

MkV model

In RevBayes this is done internally and all non varying characters will be removed before the inference

In Beast you will see the dummy characters in the xml file produced from beauti

```
<data
id="Zamora_Smith_part"
spec="Alignment"
dataType="standard">

  <sequence id="seq_Kinzercystis" spec="Sequence" taxon="Kinzercystis" totalcount="6"
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  <sequence id="seq_Gogia" spec="Sequence" taxon="Gogia" totalcount="6"
    value="000500010?200100--001001000 012345" />
  <sequence id="seq_Sinoecrinus" spec="Sequence" taxon="Sinoecrinus" totalcount="6"
    value="002500010?200100--0?1001000 012345" />
  <sequence id="seq_Akadocrinus" spec="Sequence" taxon="Akadocrinus" totalcount="6"
    value="00?5?0010?200100?-0???01011 012345" />
  <sequence id="seq_Ridersia" spec="Sequence" taxon="Ridersia" totalcount="6"
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```

MkV model

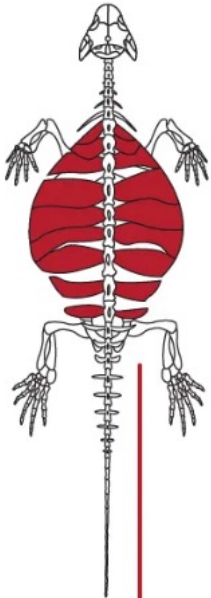


	True Branch Length	Mk	Mkv
Percent correct	-	74.0	99.8
Branch A	0.2	241,750 (±349,100)	0.206 (±0.060)
Branch B	0.05	0.43210 (±0.13756)	0.050 (± 0.018)
Branch X	0.05	54.646 (±1,725.3)	0.052 (± 0.023)
Branch C	0.2	143,950 (±228,910)	0.206 (± 0.059)
Branch D	0.05	0.022 (± 0.054)	0.051 (±0.019)

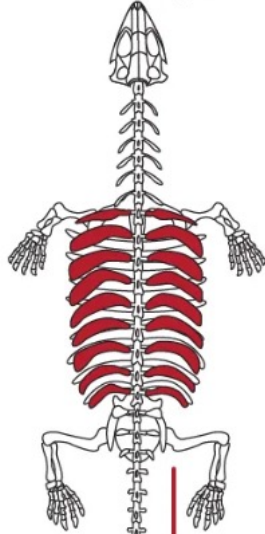
Among-character rate variation

Turtle shell evolution

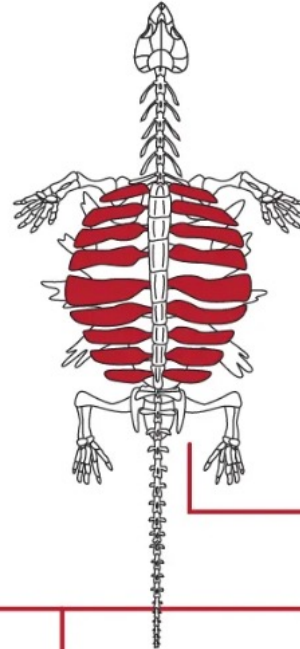
Eunotosaurus
~260 mya



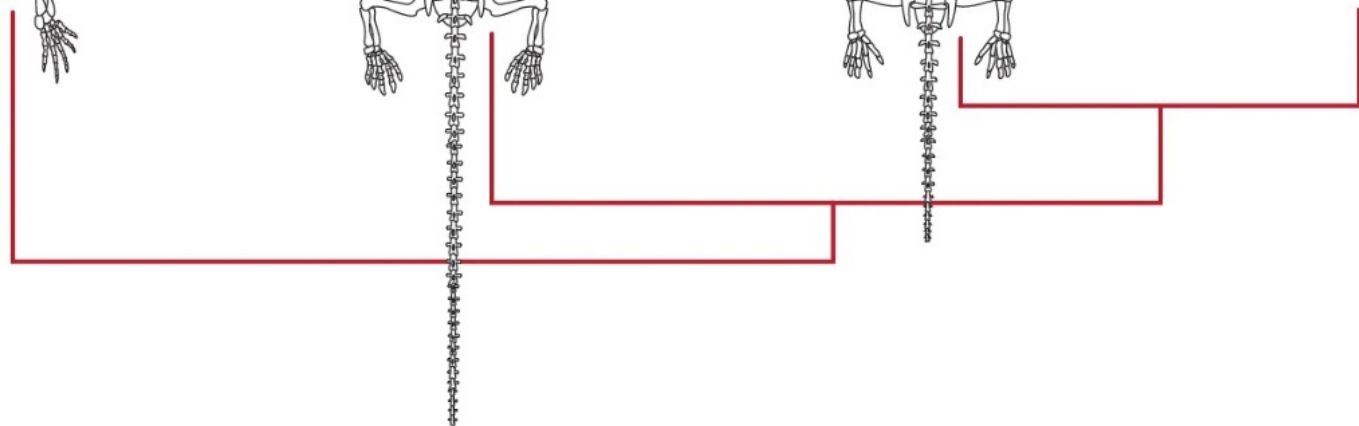
Pappochelys
~240 mya



Odontochelys
~220 mya



Proganochelys
~210 mya



Among-character rate variation

	T1	T2
Taxa A	0	0
Taxa B	0	1
Taxa C	1	2

The transition rate
will impact branch
lengths

Slow rate of evolution



Fast rate of evolution

Relative to each other!

Among-character rate variation

What do we do?

	T1	T2
Taxa A	0	0
Taxa B	0	1
Taxa C	1	2

Allow these traits to evolve at different rates:

- Specify which traits evolve fast
- Use a gamma model to account for rate heterogeneity

Among-character rate variation

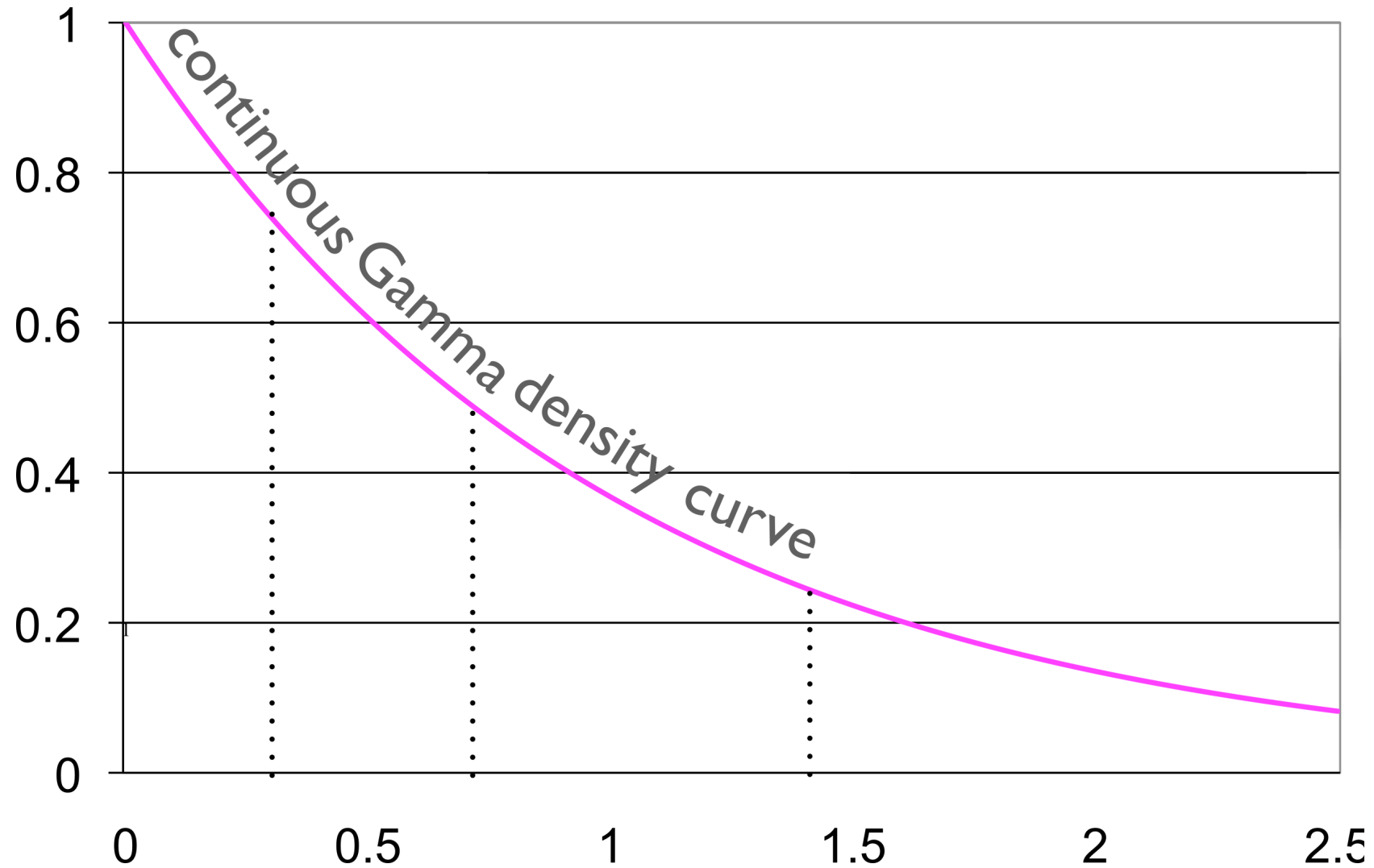
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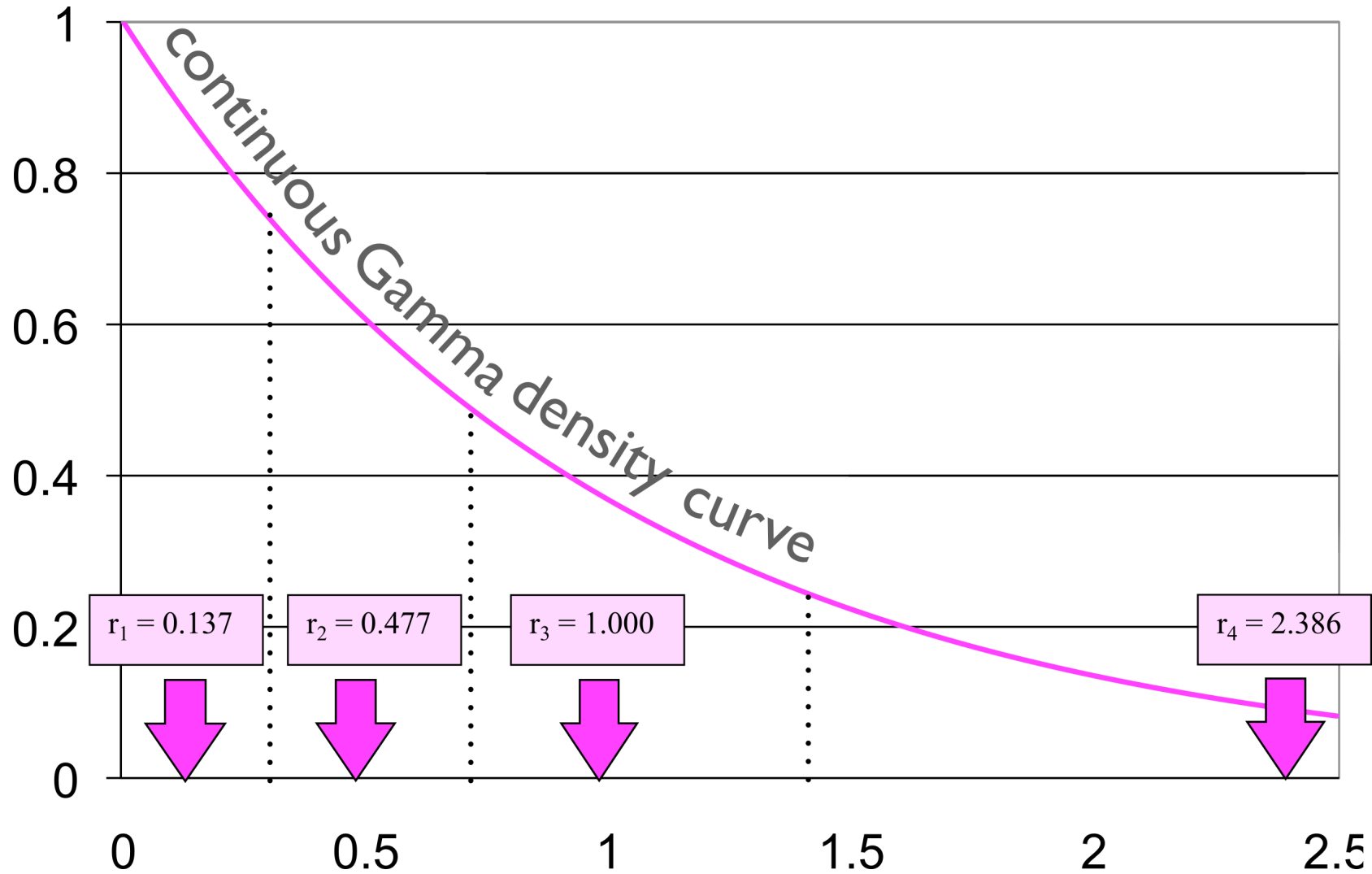
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Mk(V) + Gamma



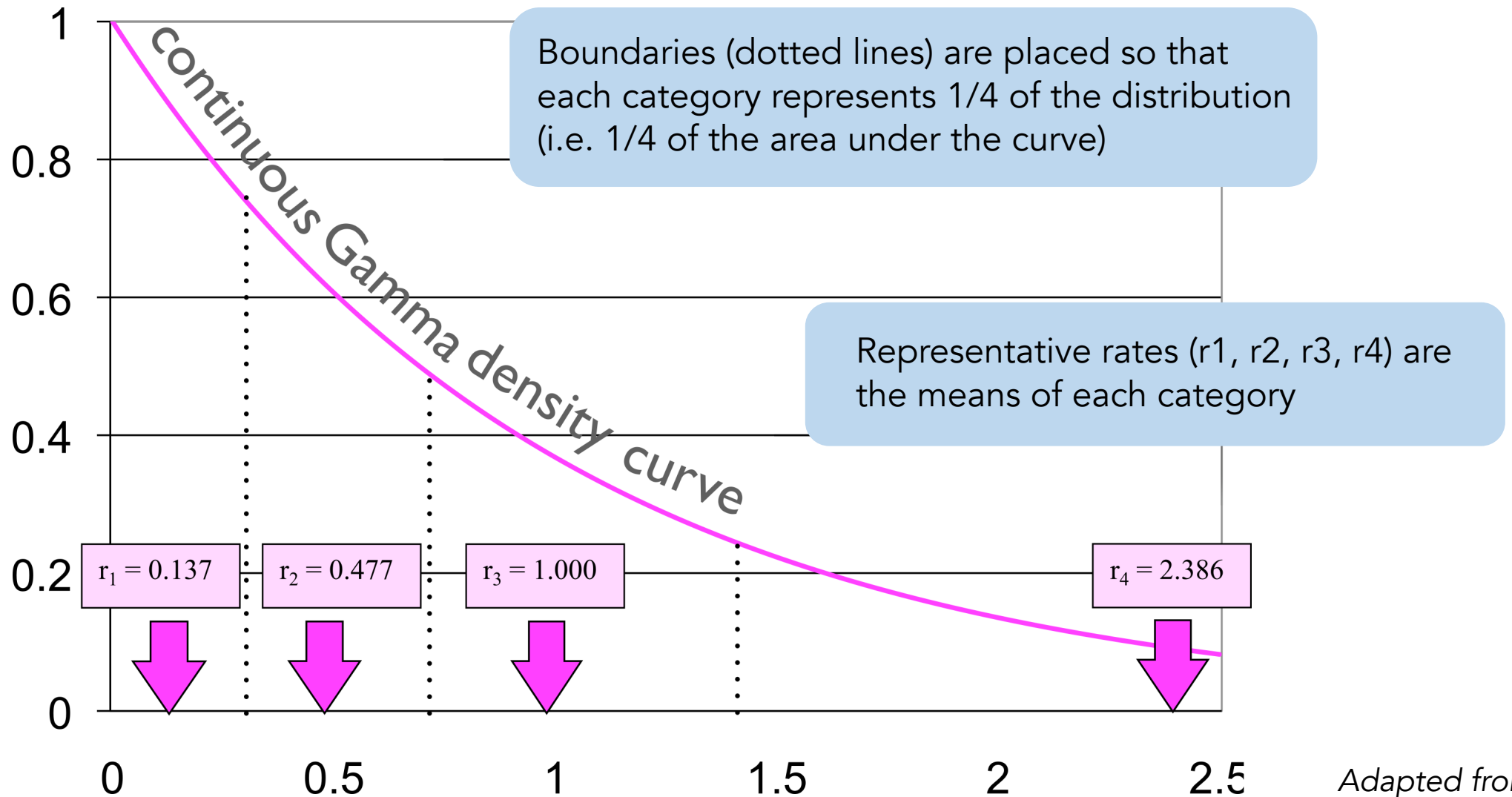
Adapted from Paul
Lewis PhyloSeminar

Mk(V) + Gamma



Adapted from Paul
Lewis PhyloSeminar

Mk(V) + Gamma



Mk(V) + Gamma

What do we do?

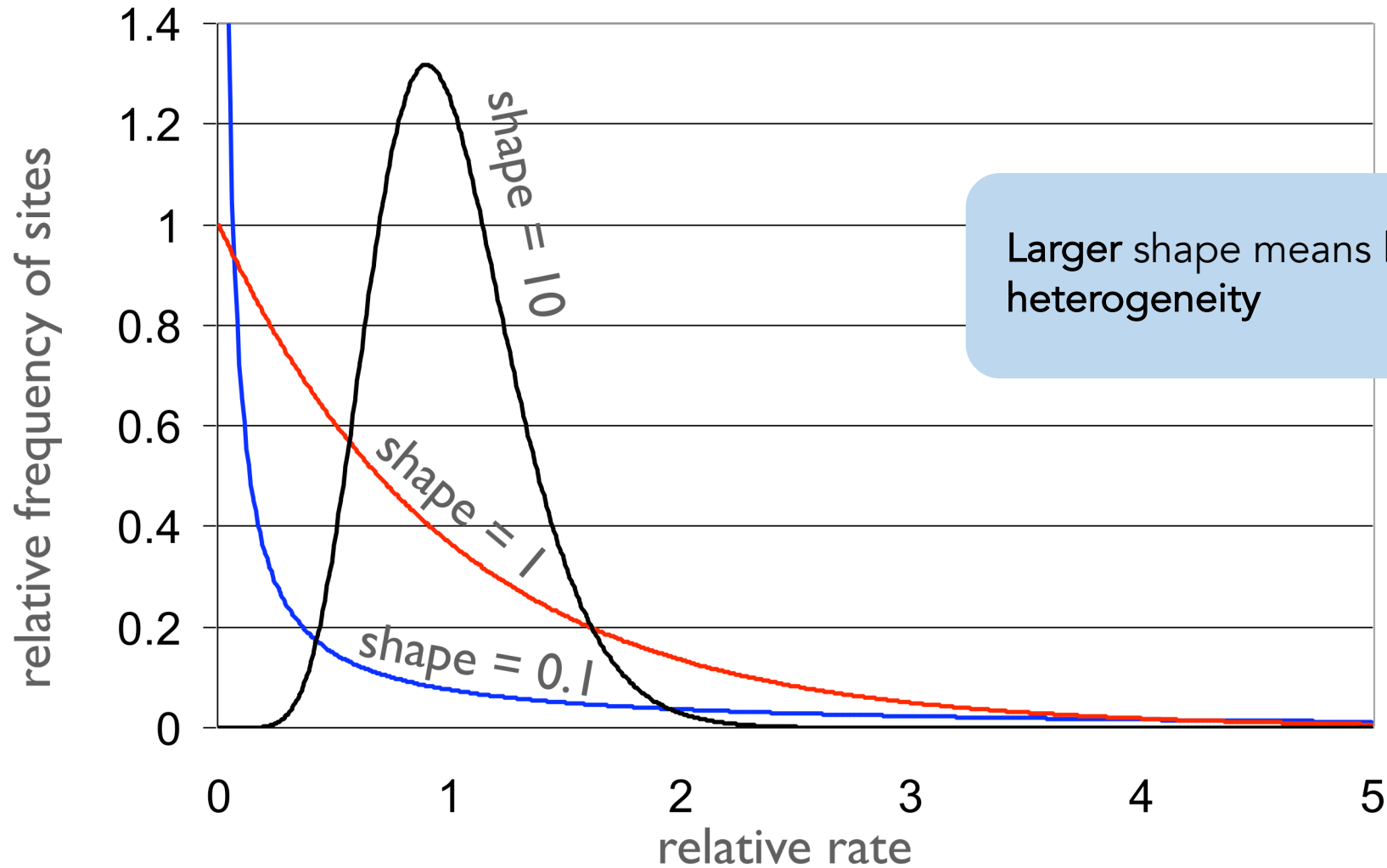
	T1	T2	
Taxa A	0	0	
Taxa B	0	1	
Taxa C	1	2	Faster R (R4)

Slower R (R1,2)

Allow each trait to evolve according to the rates drawn from the gamma distribution

One rate will fit the best and be the most influential for the likelihood calculation

Mk(V) + Gamma



Partitioning Data

Grouping together parts of the alignment that have similar characteristics and or may have **evolved together** due to evolutionary pressures

The **defaults** in many phylogenetic software is to group by maximum observed state size

$$Q = \begin{pmatrix} -\mu_0 & \mu_{01} & \mu_{02} & \mu_{03} \\ \mu_{10} & -\mu_1 & \mu_{12} & \mu_{13} \\ \mu_{20} & \mu_{21} & -\mu_2 & \mu_{23} \\ \mu_{30} & \mu_{31} & \mu_{32} & -\mu_3 \end{pmatrix},$$

$$\begin{bmatrix} -\mu_0 & \mu_{01} \\ \mu_{10} & -\mu_1 \end{bmatrix} \quad \begin{bmatrix} -\mu_0 & \mu_{01} & \mu_{02} \\ \mu_{10} & -\mu_1 & \mu_{12} \\ \mu_{20} & \mu_{21} & -\mu_2 \end{bmatrix}$$

Partitioning Data

When should we partition our data?

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If we have presence (1) absence (0) traits partitioning will always be a logical approach: what would transitioning to state 2 in this scenario even mean?

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When should we partition our data?

If we have presence (1) absence (0) traits partitioning will always be a logical approach: what would transitioning to state 2 in this scenario even mean?

We should be cautious for traits describing a trait – just because we do not observe a state 2 can we be absolutely certain there never was one?

Justifying partitioning schemes is very important as they have a major impact on inference results

Other morphological models

Ordered Characters

Ordered characters can be placed in an order so that transitions only occur between adjacent states.



For example, “**intermediate**” species that are somewhere in between limbed and limbless – for example, the “mermaid skinks” (*Sirenoscincus*) from Madagascar, so called because they lack hind limbs. An ordered model might only allow transitions between limbless and intermediate, and intermediate and limbed; it would be impossible under such a model to go directly from limbed to limbless without first becoming intermediate.

For unordered characters, any state can change into any other state.

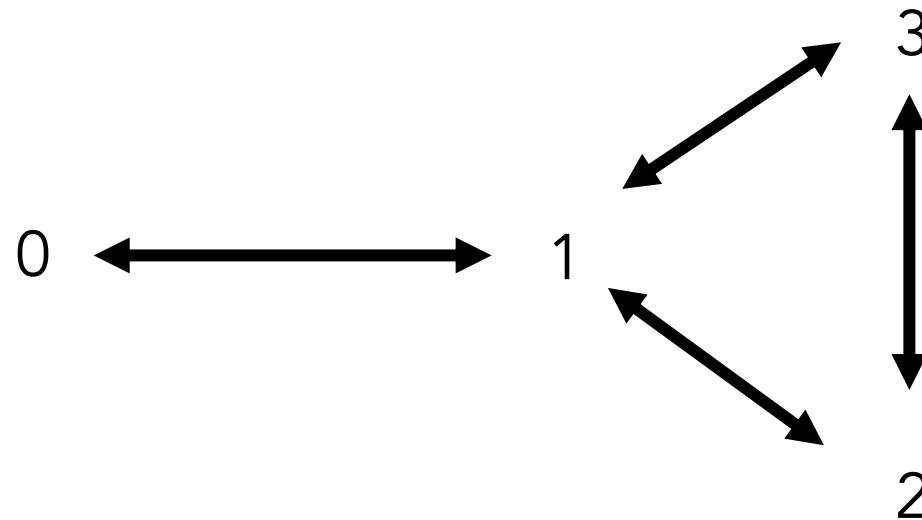
Ordered Characters



All characters ordered:



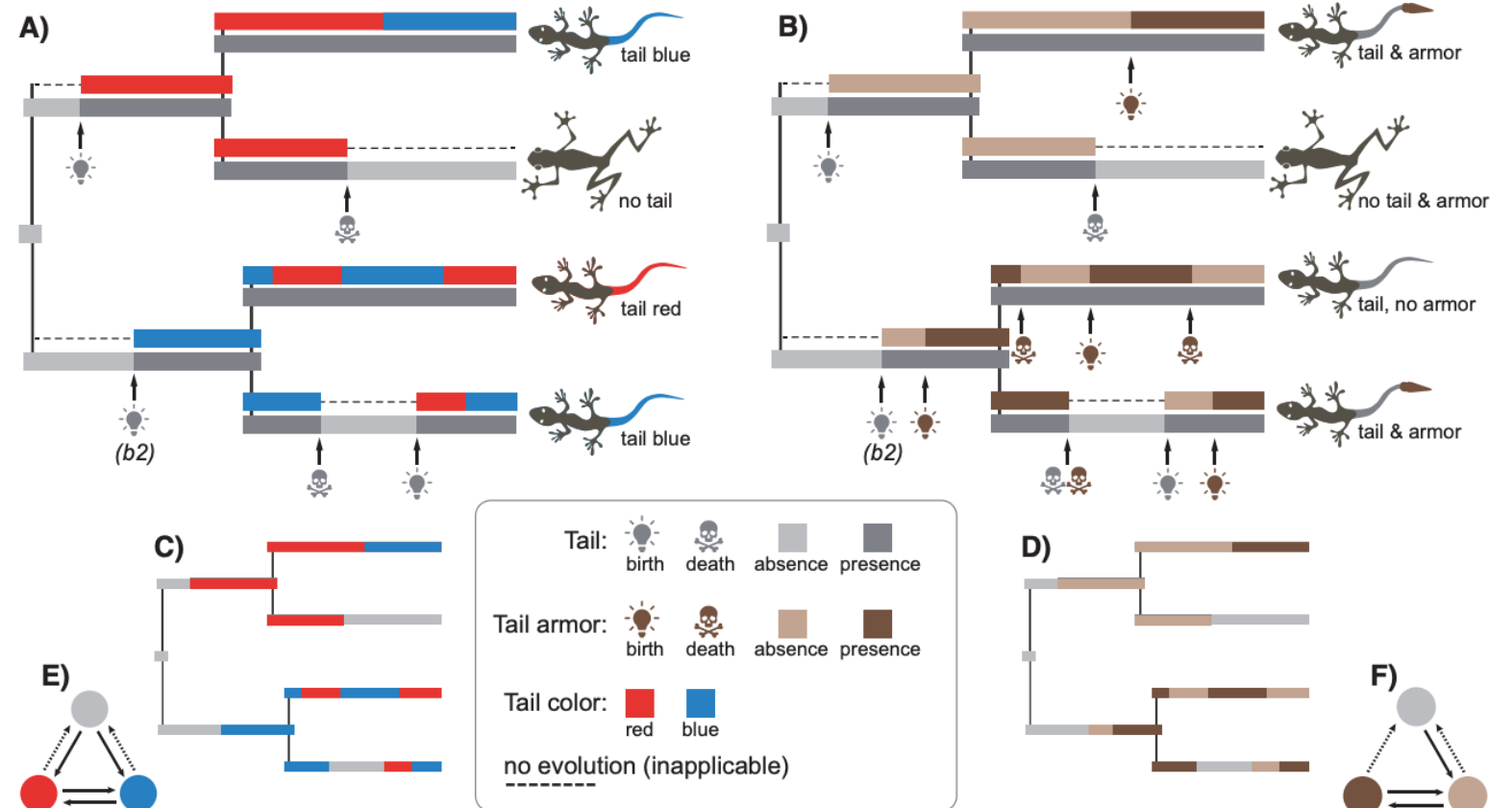
Specific characters
ordered:



Embedded dependency model

Markov models for
phylogenetic inference
with anatomically
dependent (inapplicable)
morphological characters

Non-applicable characters
only considered when
they are present (1)

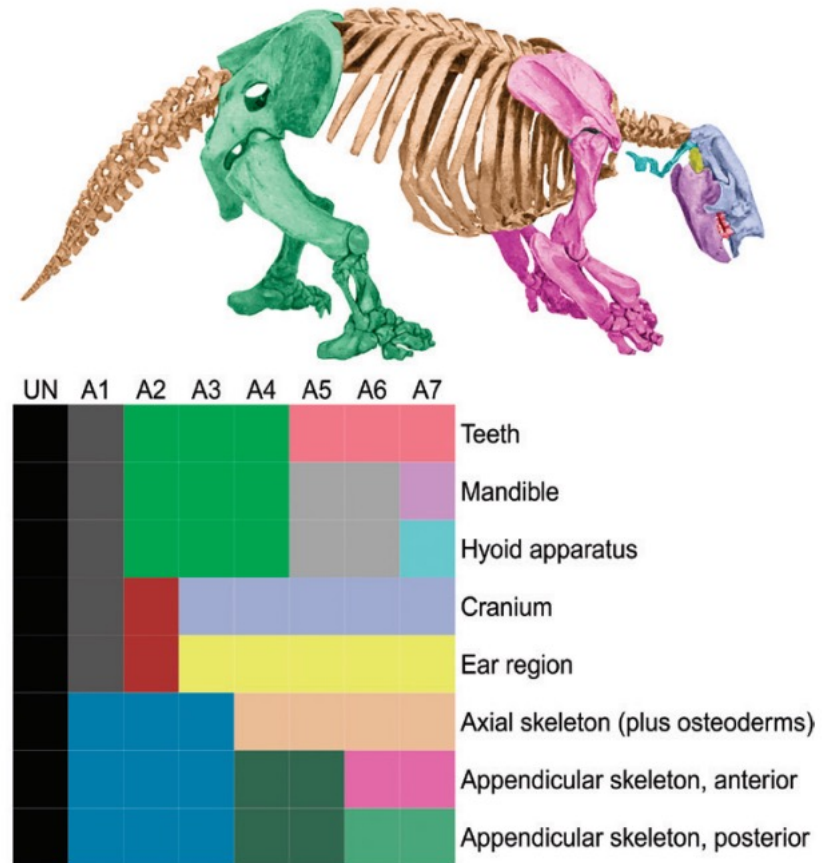


Alternative Partitioning schemes

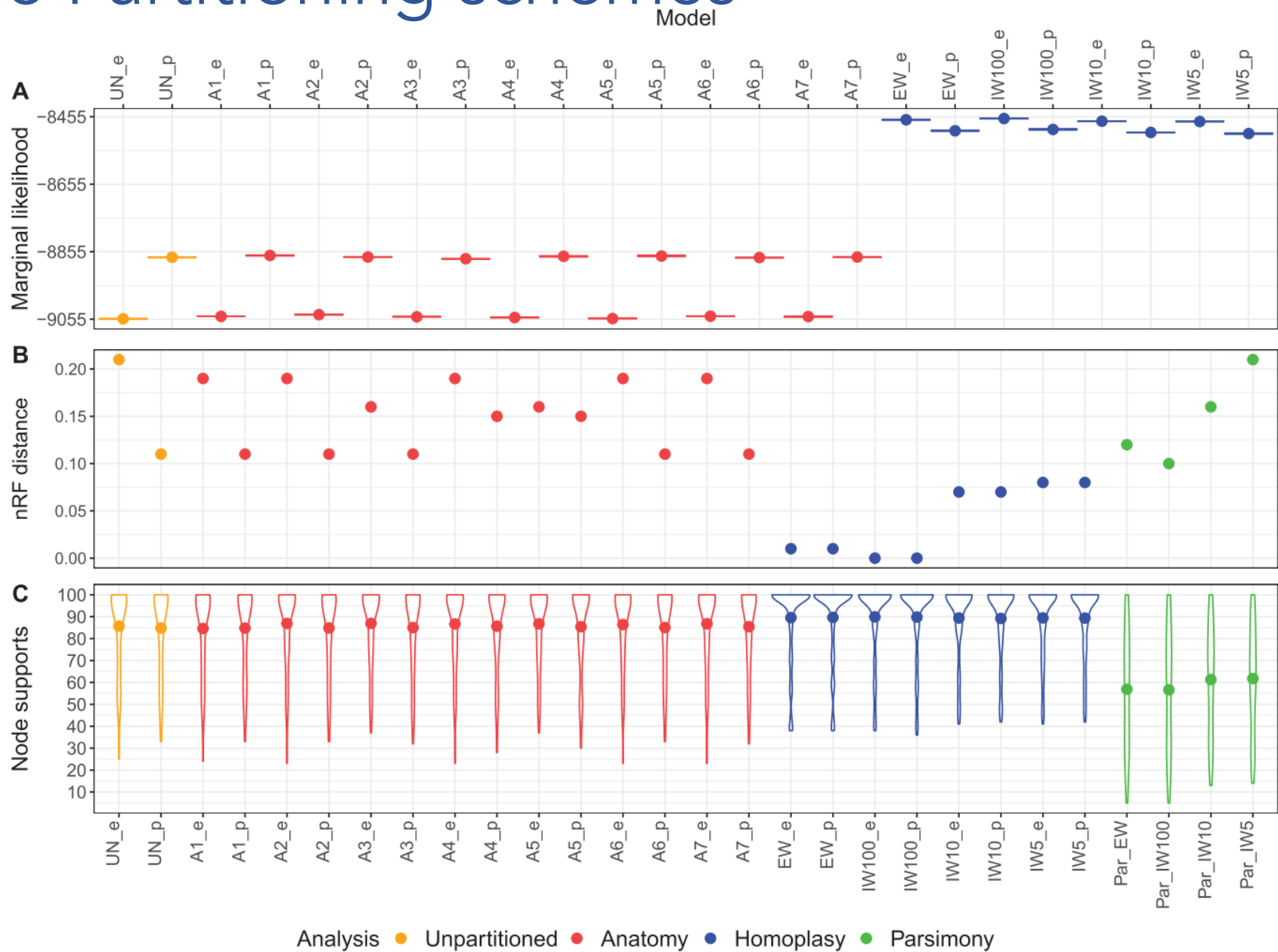
Reassessing the phylogeny and divergence times of sloths (Mammalia: Pilosa: Folivora)

Characters can be groups based on anatomical region

Other criteria such as the degree of homoplasy present in a character was explored in this study – and found to be a better fit using Bayes factors



Alternative Partitioning schemes



Challenges with morphological data

Generalising assumptions across different traits is often not possible

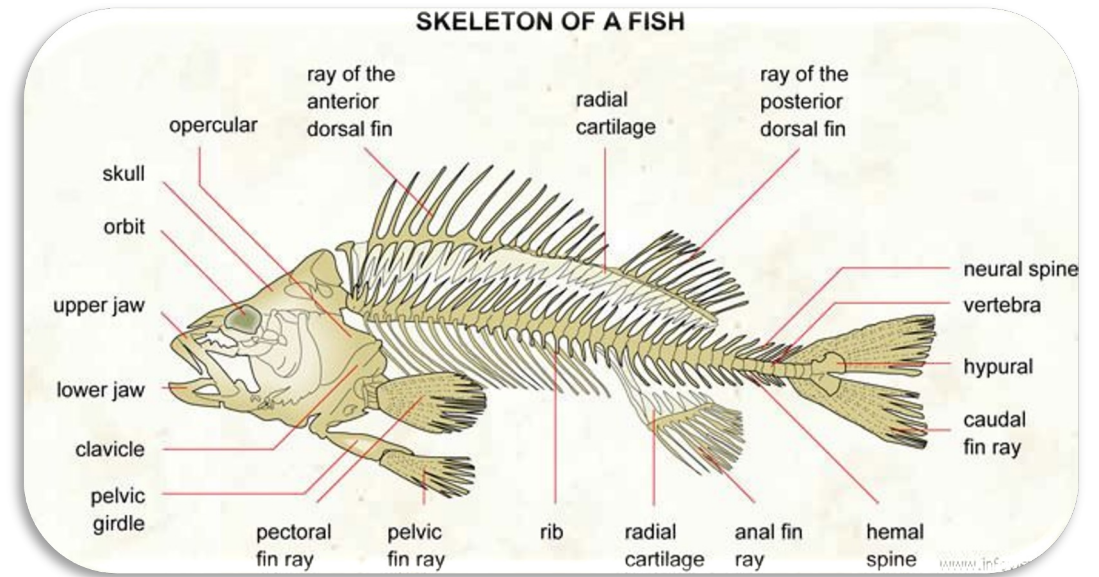
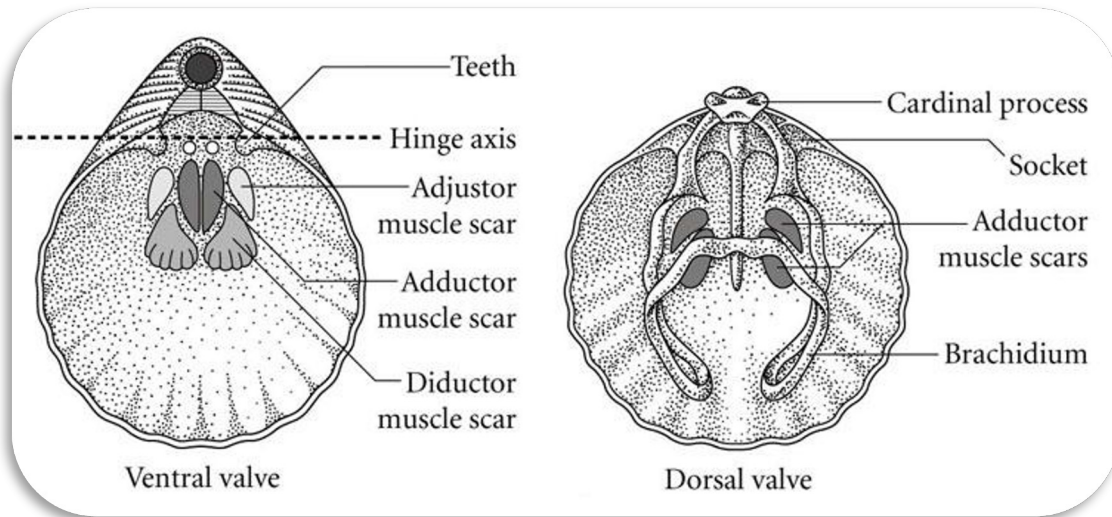
Modelling special characters in matrices

```
001510010?00-100--00000000000
000500010?200100--00100100000
002500010?200100--0?100100000
00?5?0010?200100?-0???010110
0015000101201000430100011111
0015000101201010440111011111
??050?????201000440?11011111
```

Challenges with morphological data

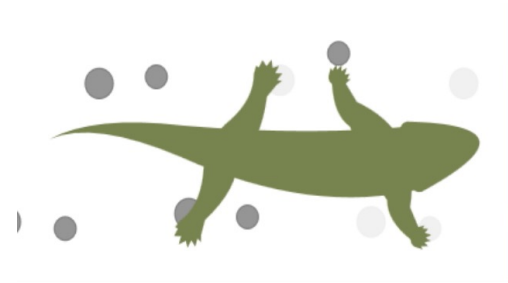
Morphological matrices are often quite small:

- Collection is very time consuming
- Number of characters available can be very small depending on the group



Model impact on key parameter estimates

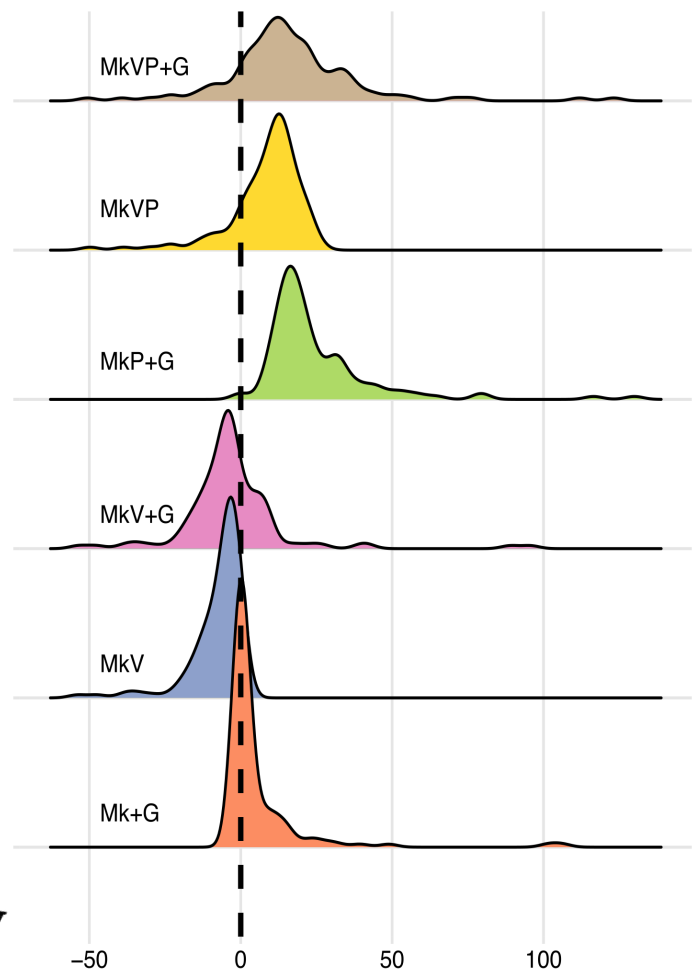
Example of 114 empirical **tetrapod** matrices



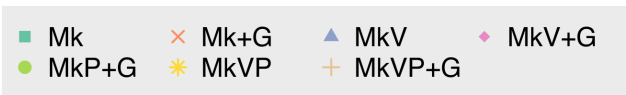
Looked at the impact on:

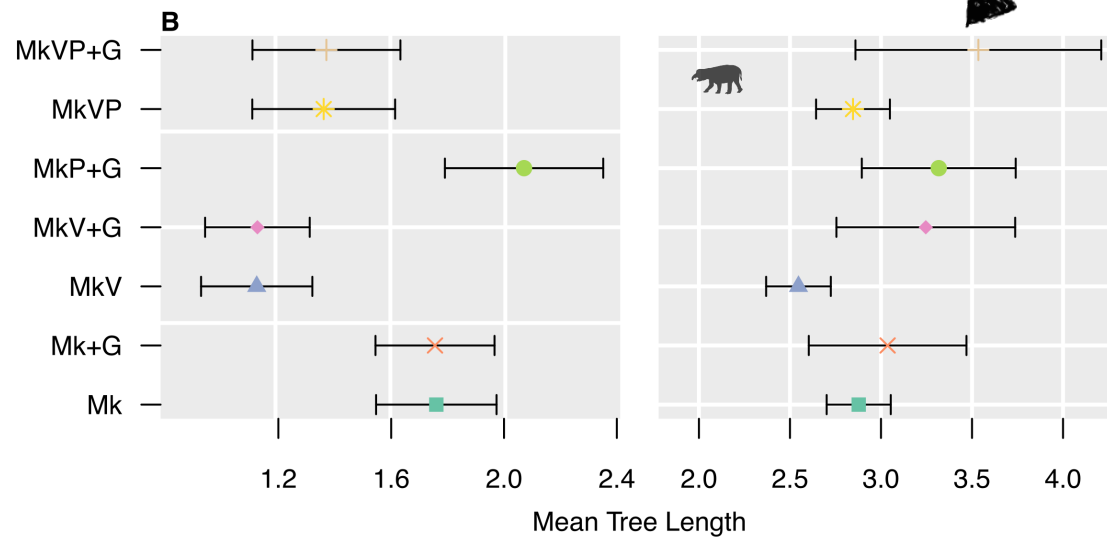
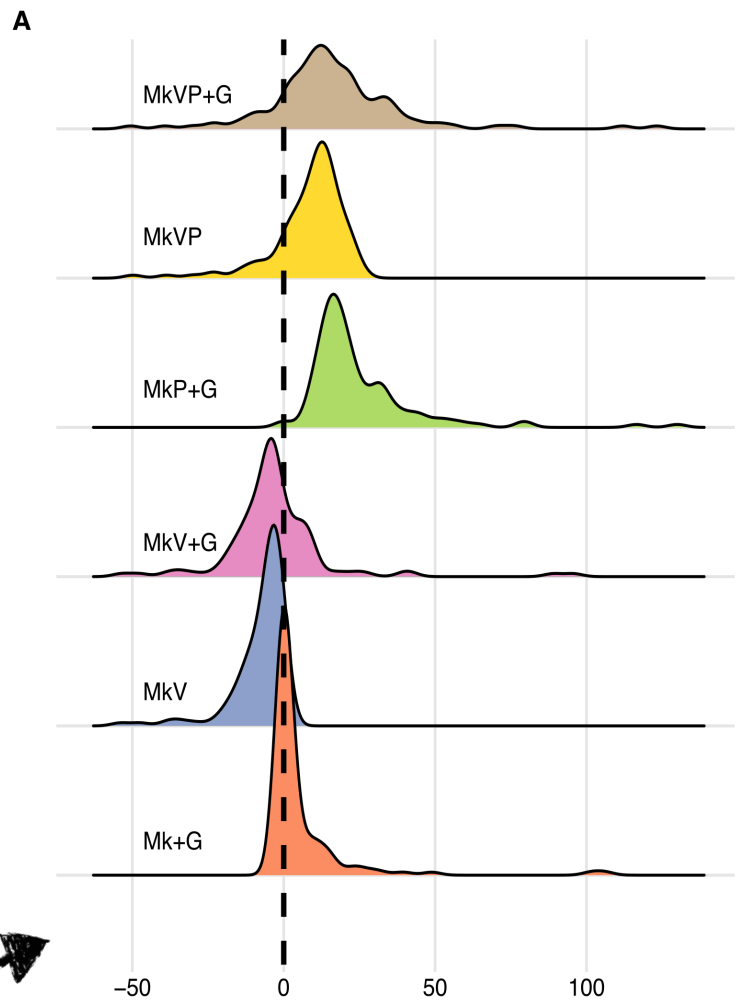
- branch lengths (**evolutionary distances**)
- Tree topology (**species relationships**)

A

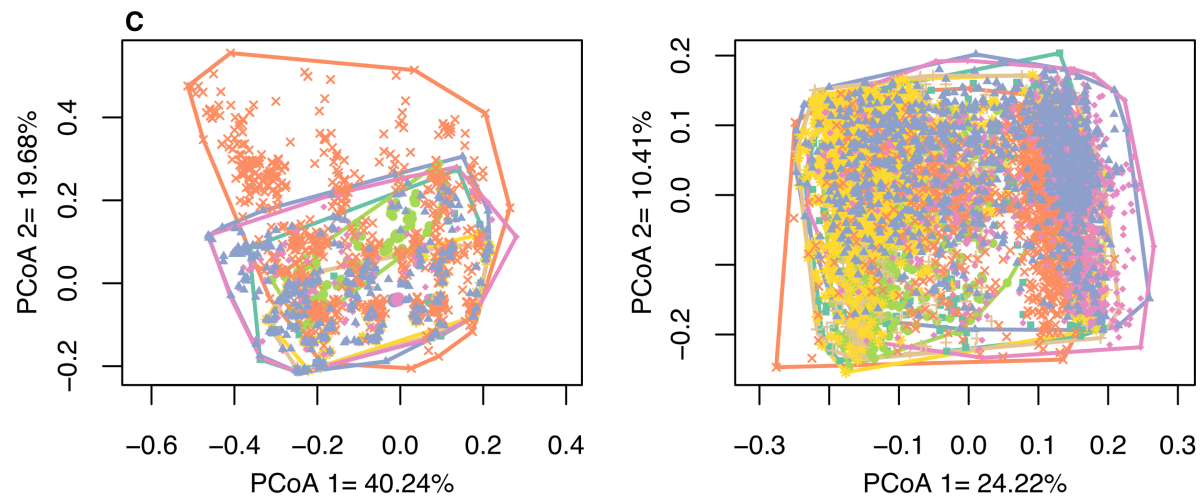


Percentage difference in
tree length relative to Mk
model





Tree length of
two different
data sets



Rf
distances
of two
data sets

Percentage difference in
tree length relative to Mk
model



How do we choose a
model?

Model selection

Bayes factors are commonly used to determine the **relative fit** between model.

It relies on comparing the marginal likelihoods approximated from different models.

The ML measures the average fit of a model to our data.

We use MCMC to avoid calculating this number as it is computationally expensive and often not directly possible.

Model selection

$$P(\text{model} \mid \text{data}) = \frac{P(\text{data} \mid \text{model}) P(\text{model})}{P(\text{data} \mid \text{model})}$$

Diagram illustrating the components of the model selection equation:

- Posterior**: Points to $P(\text{model} \mid \text{data})$
- Likelihood**: Points to $P(\text{data} \mid \text{model})$ in the numerator
- Priors**: Points to $P(\text{model})$ in the numerator
- Marginal likelihood**: Points to $P(\text{data} \mid \text{model})$ in the denominator

Marginal likelihood

Marginal probability of the data (denominator in Bayes' rule) is the expected value of the likelihood with respect to the prior distribution.

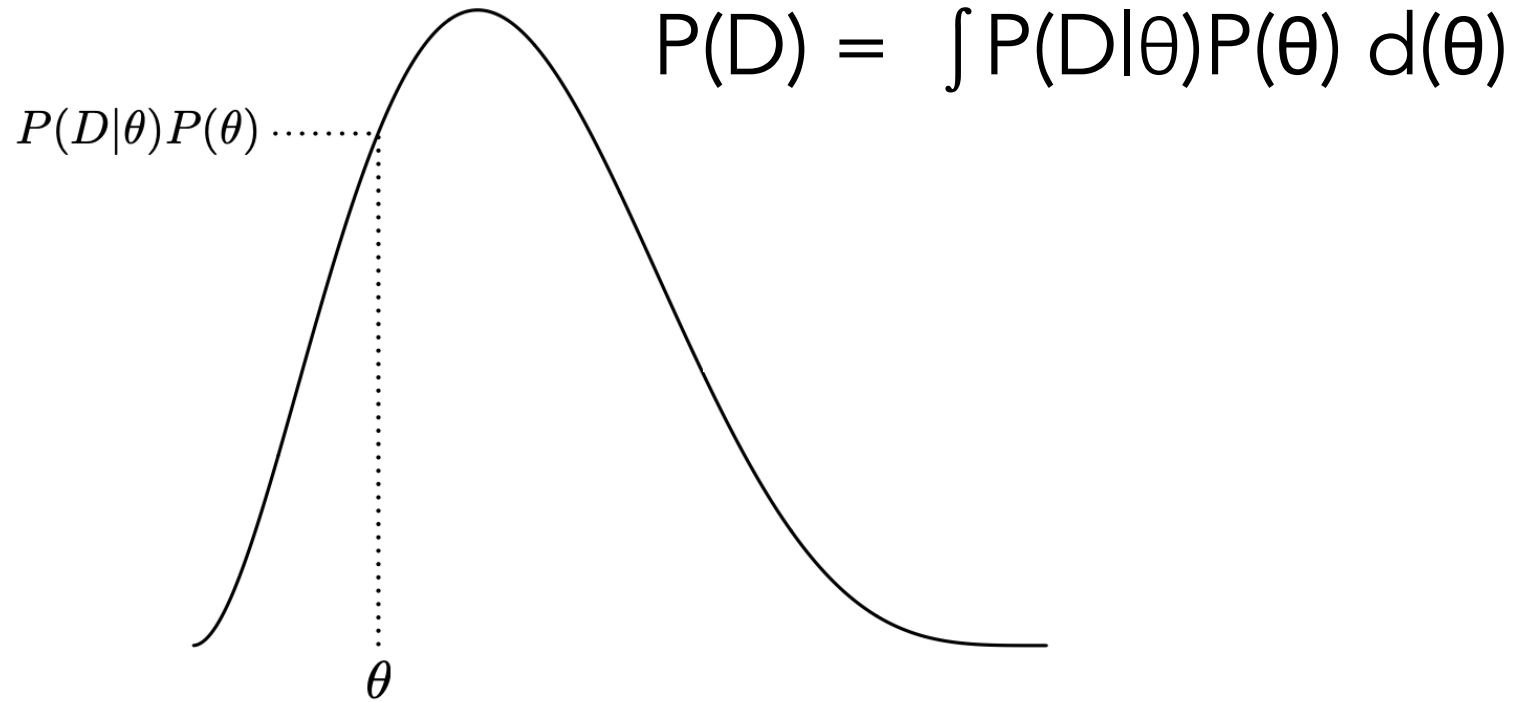
If likelihood measures model fit, then the marginal likelihood measures the average fit of the model to the data over all parameter values.

What is the expected value?

Marginal likelihood

$P(\text{data} \mid \text{model})$

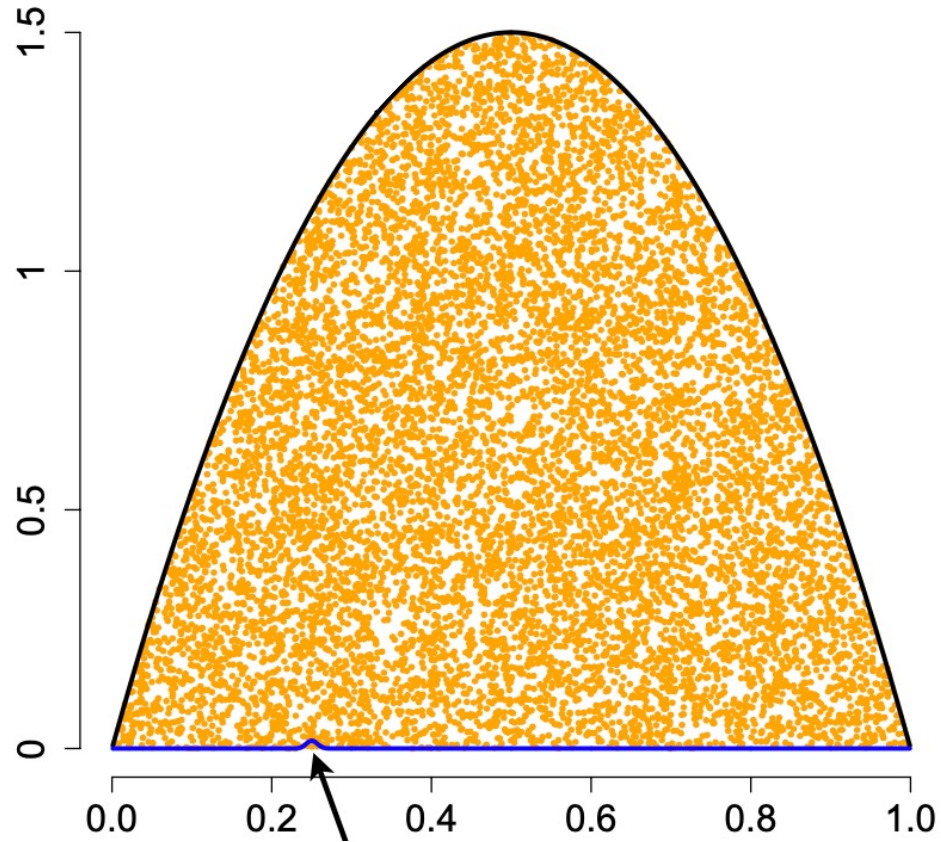
The marginal likelihood is used to evaluate the overall fit of the model to the data, integrating over all parameter values.



Marginal likelihood

$P(\text{data} \mid \text{model})$

Very small, single
number between the
posterior distribution
and the prior



Approximating the marginal likelihood

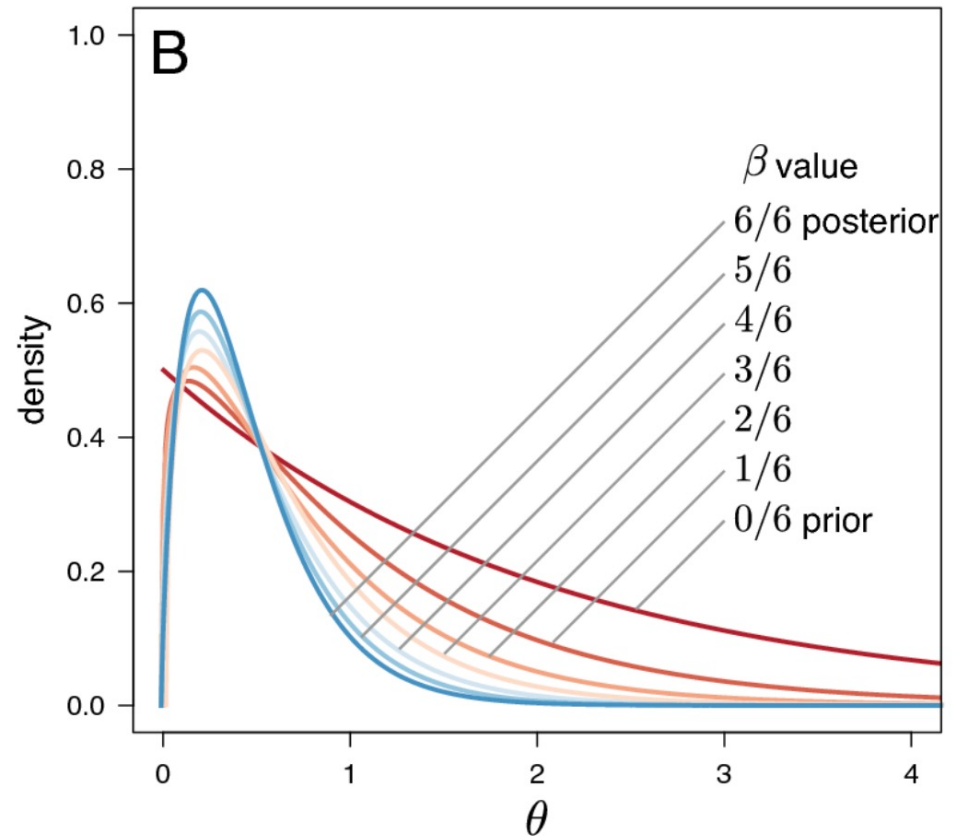
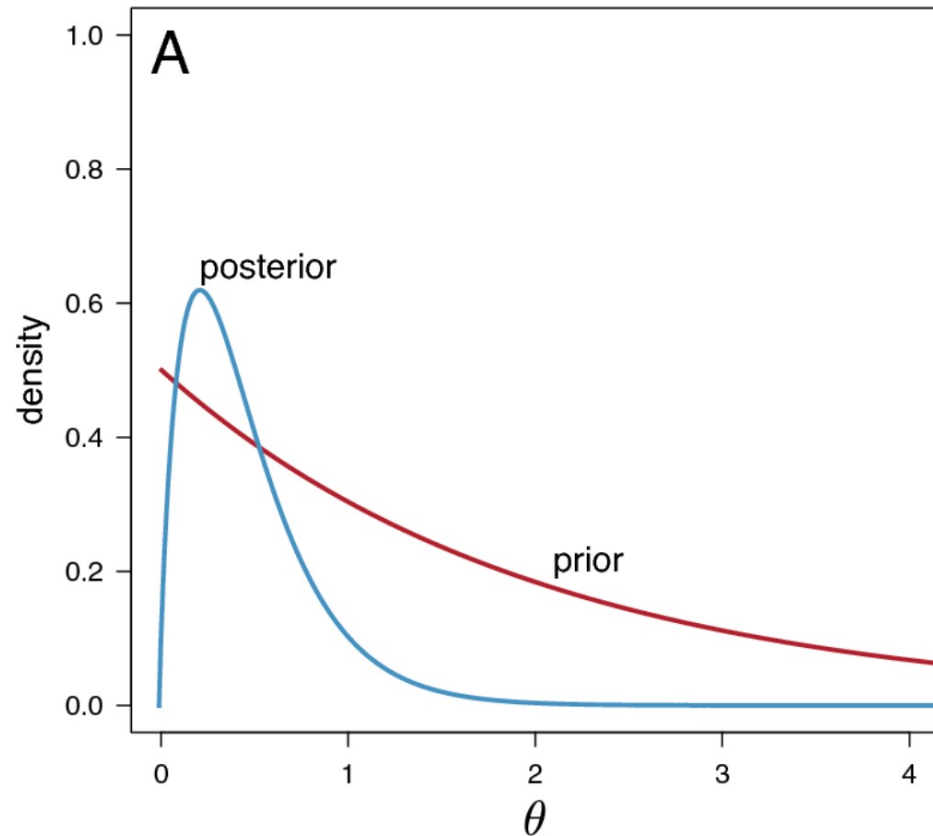
There are two common algorithms to do this:

- Stepping stone
- Path sampling

Both of these approaches are computationally expensive

Stepping-stone algorithms are like a series of MCMC simulations that iteratively sample from a specified number of distributions that are discrete steps between the posterior and the prior probability distributions.

Stepping stone algorithm



$\beta = 1$ this is the posterior distribution

$\beta = 0$ this is the prior distribution

Bayes factors

$$B_{01} = \frac{P(D | M_0)}{P(D | M_1)} = \frac{\text{Marginal likelihood for model } M_0}{\text{Marginal likelihood for model } M_1}$$

Bayes factors

$$B_{01} = \frac{P(D | M_0)}{P(D | M_1)} = \frac{\text{Marginal likelihood for model } M_0}{\text{Marginal likelihood for model } M_1}$$

Marginal likelihoods are often on the log scale so the Bayes factor can be calculated as:

$$\log B_{01} = \log P(D | M_0) - \log P(D | M_1)$$

Interpreting Bayes factors

Strength of evidence	$BF(M0, M1)$	$\log(BF(M0, M1))$
Negative (supports M_1)	<1	<0
Barely worth mentioning	1 to 3.2	0 to 1.16
Substantial	3.2 to 10	1.16 to 2.3
Strong	10 to 100	2.3 to 4.6
Decisive	>100	>4.6

Exercise:

Use stepping stone to
determine the fit
between models

Issues with Bayes factors

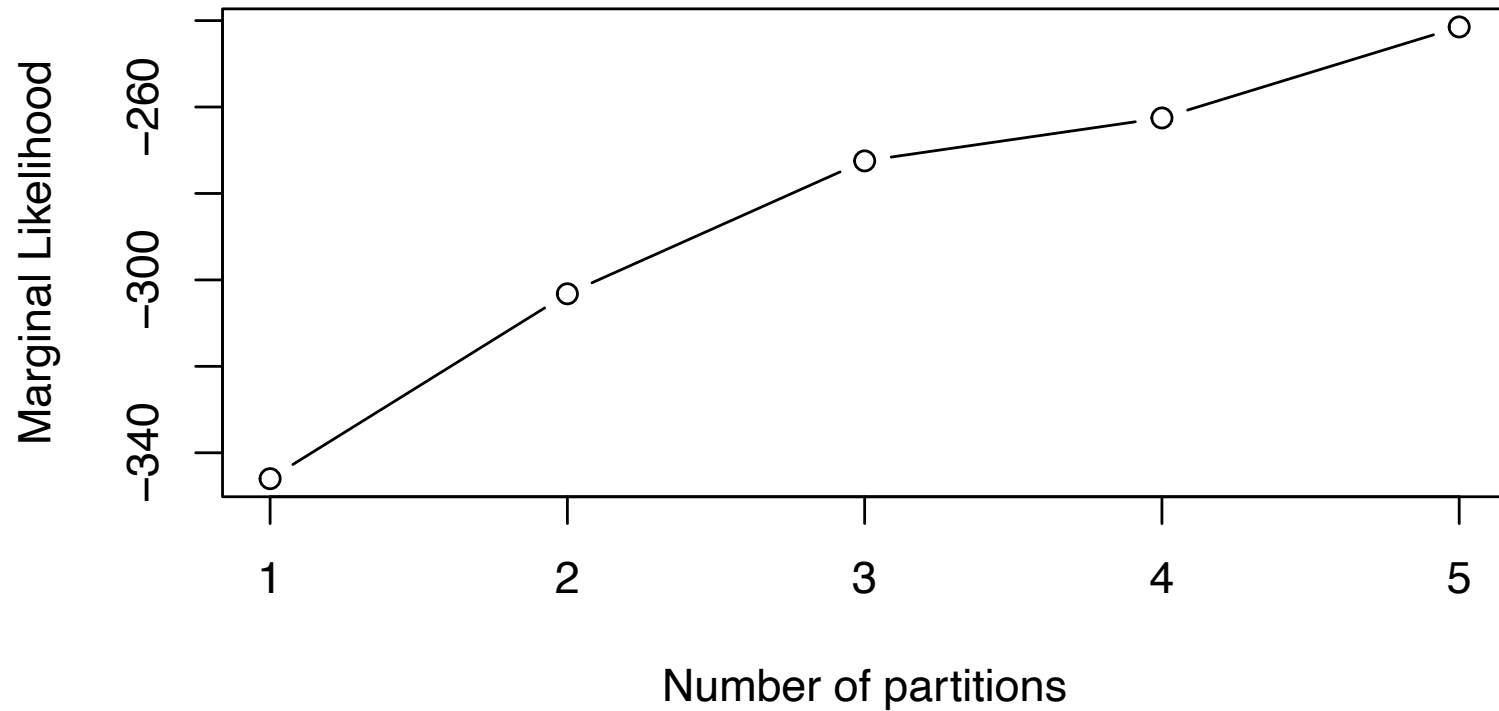
The way we **partition data** for morphological data is different to molecular

0	1	0	0	2	3
2	0	1	1	0	2
1	1	2	1	3	1

Unpartitioned everything
in Q-matrix of size 4

Partitioning the data puts
characters into correctly
sized Q-matrix

Issues with Bayes factors



Data set with 6 states

As the number of partitions increases, so does the likelihood

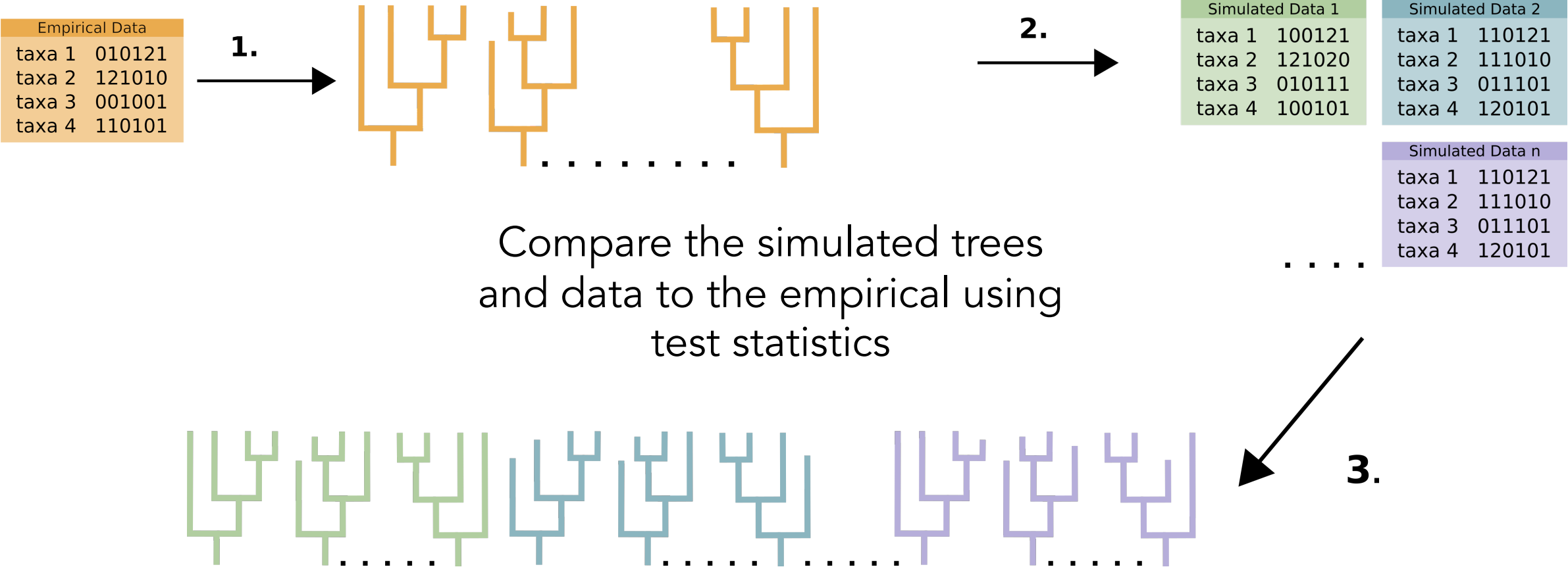
Model Adequacy

Assess whether a model is capturing the evolutionary dynamics that generated the data

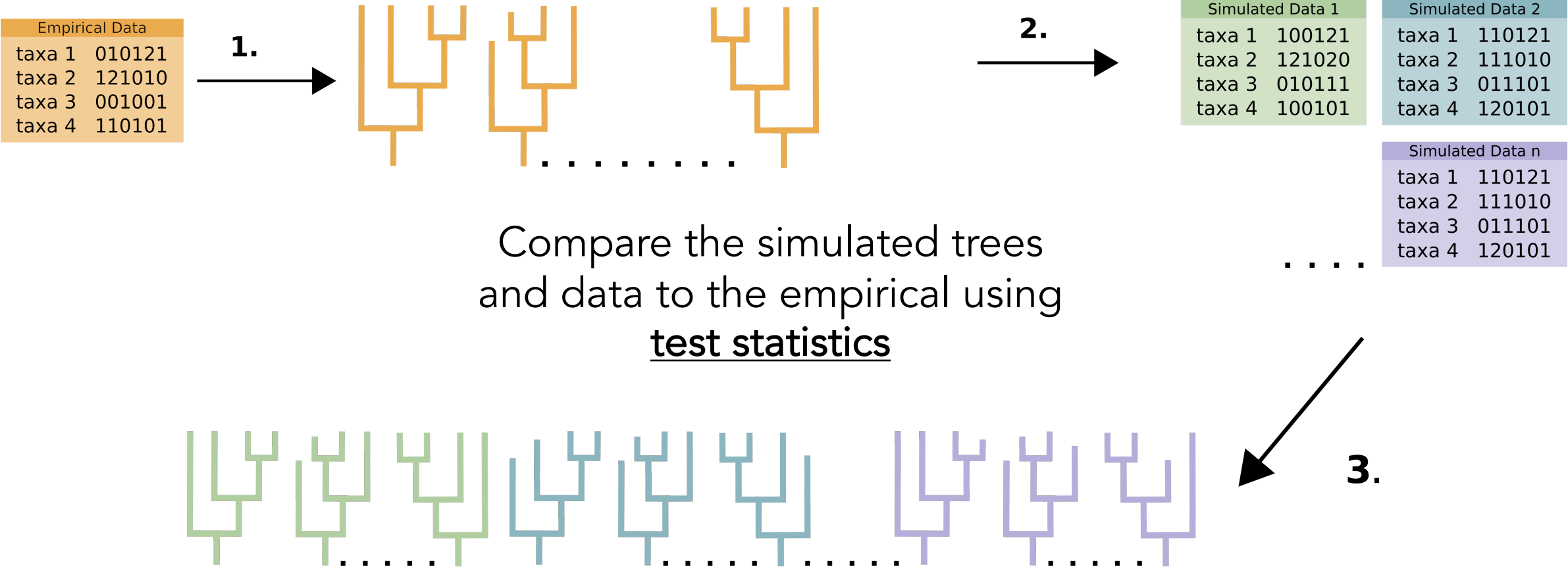
Gives the **absolute fit**

One approach is **Posterior Predictive Simulations**

Posterior Predictive Simulations



Posterior Predictive Simulations



Test Statistics

A test statistic is a **numerical summary** of data.

A value that captures the characteristic of your data.

For PPS we have 3 categories:

Data-based, inference-based, mixed

Test Statistics

Empirical Data	
taxa 1	0 1 0 1 2 1
taxa 2	1 2 1 0 1 0
taxa 3	0 0 1 0 0 1
taxa 4	1 1 0 1 0 1

~

Simulated Data 1	
taxa 1	1 0 0 1 2 1
taxa 2	1 2 1 0 2 0
taxa 3	0 1 0 1 1 1
taxa 4	1 0 0 1 0 1

Simulated Data 2	
taxa 1	1 1 0 1 2 1
taxa 2	1 1 1 0 1 0
taxa 3	0 1 1 1 0 1
taxa 4	1 2 0 1 0 1

Data test stats

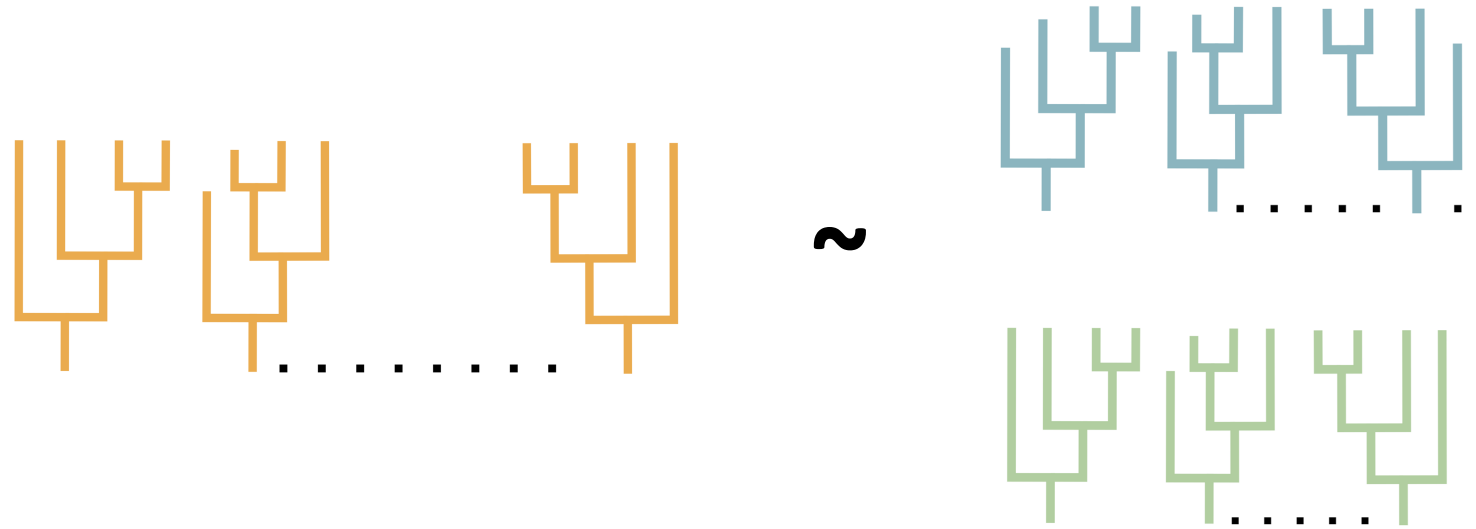
Disparity is a measure of the **range or significance of morphology** in a given sample of organisms

- i. Gowers Coefficient
- ii. Generalised Euclidean Distance

Test Statistics

Tree test stats

- i. Tree length: sum of the branch lengths.
- ii. Robinson Foulds: topological uncertainty within the posterior



Test Statistics

Mixed test stats

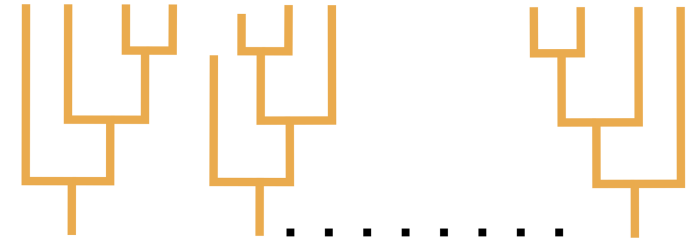
i. Consistency Index

Measures the amount of **homoplasy** (convergent evolution) in a data set

ii. Retention index

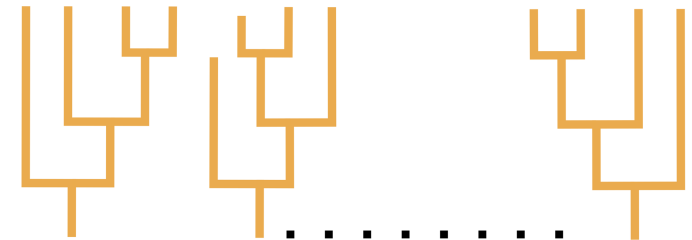
Measures the amount of **synapomorphies** in a data set

Empirical Data	
taxa 1	0 1 0 1 2 1
taxa 2	1 2 1 0 1 0
taxa 3	0 0 1 0 0 1
taxa 4	1 1 0 1 0 1



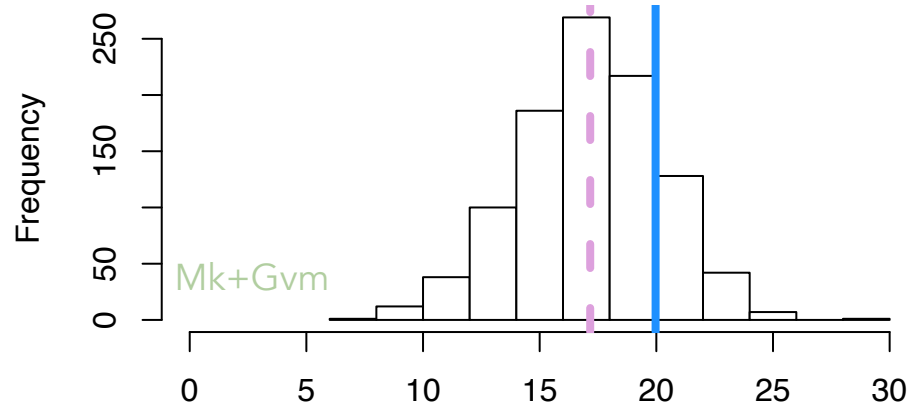
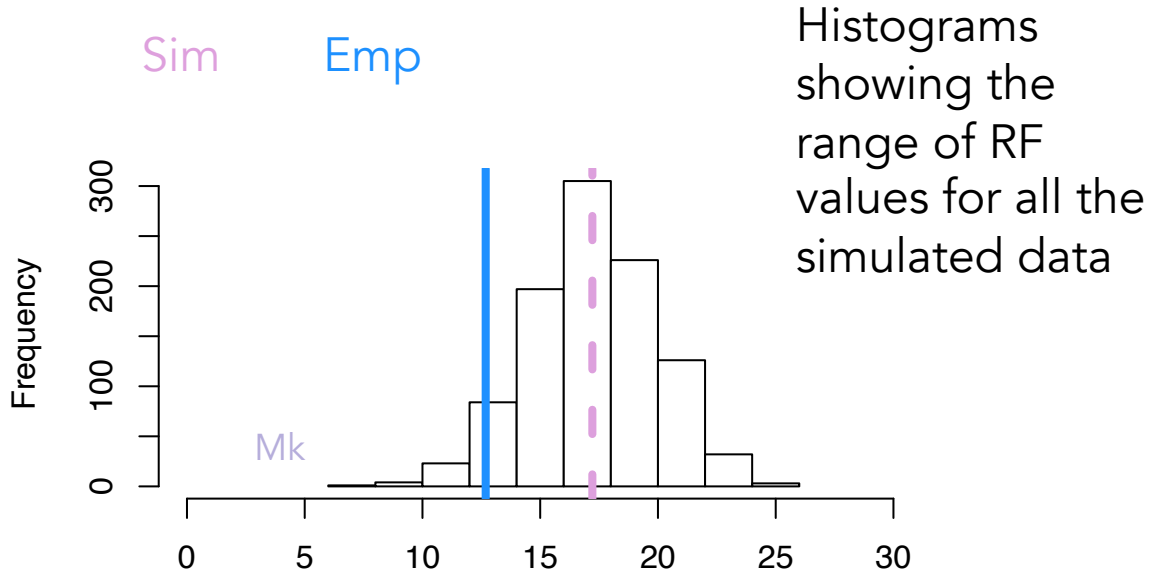
~

Simulated Data 1	
taxa 1	1 0 0 1 2 1
taxa 2	1 2 1 0 2 0
taxa 3	0 1 0 1 1 1
taxa 4	1 0 0 1 0 1



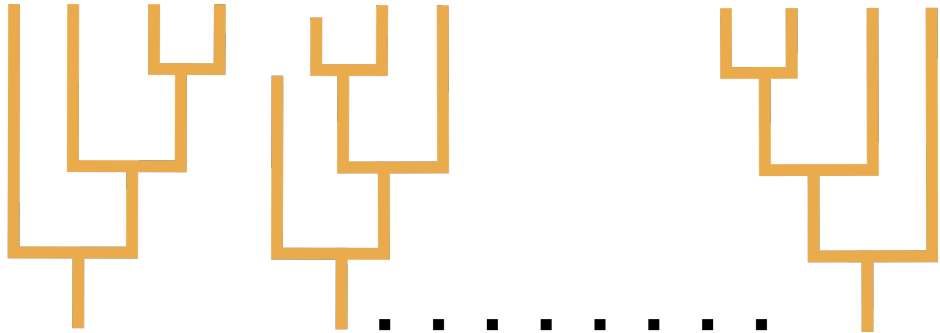
Test Statistics

Robinson Foulds Distance

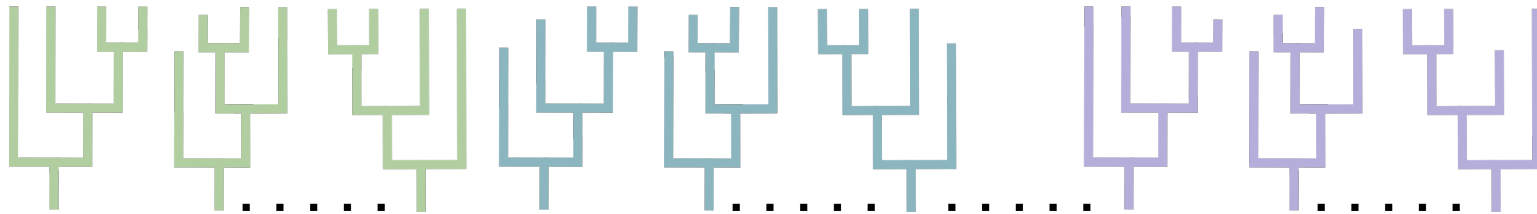


Test Statistics

Take Robinson Foulds Distance



1 empirical RF
value



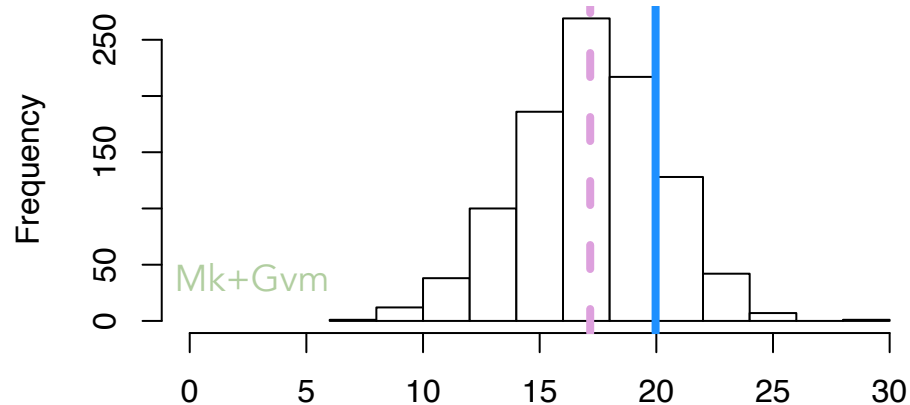
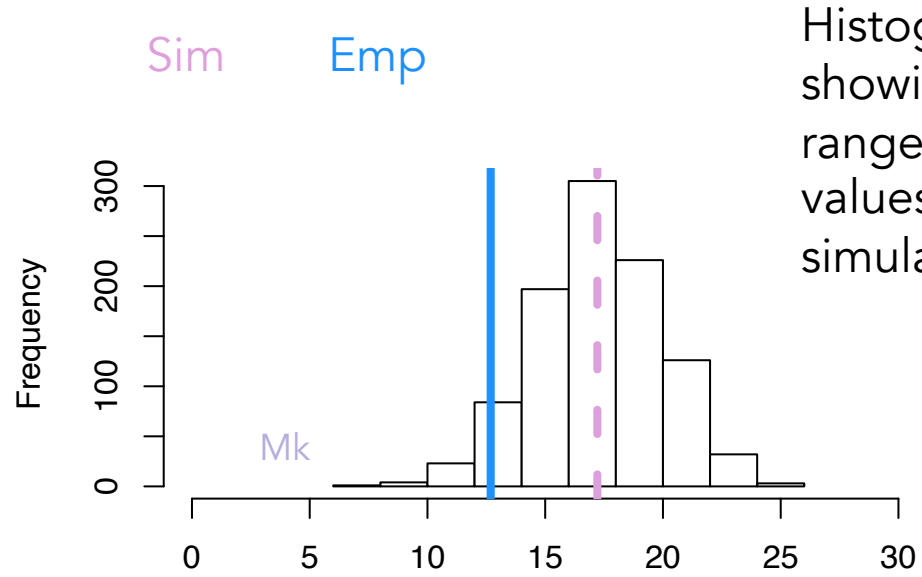
Sim 1 RF value

Sim 1 RF value

Sim n RF value

Test Statistics

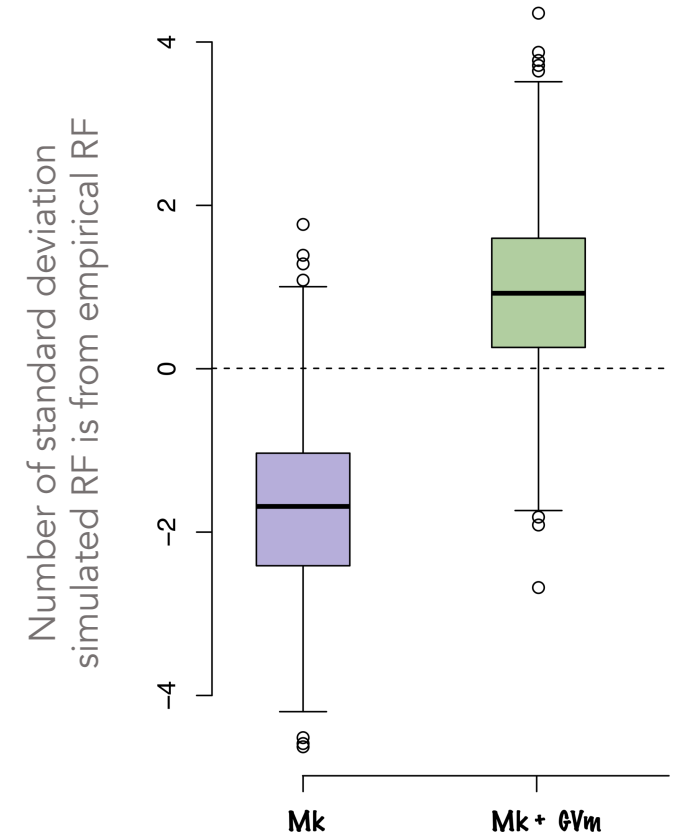
Robinson Foulds Distance



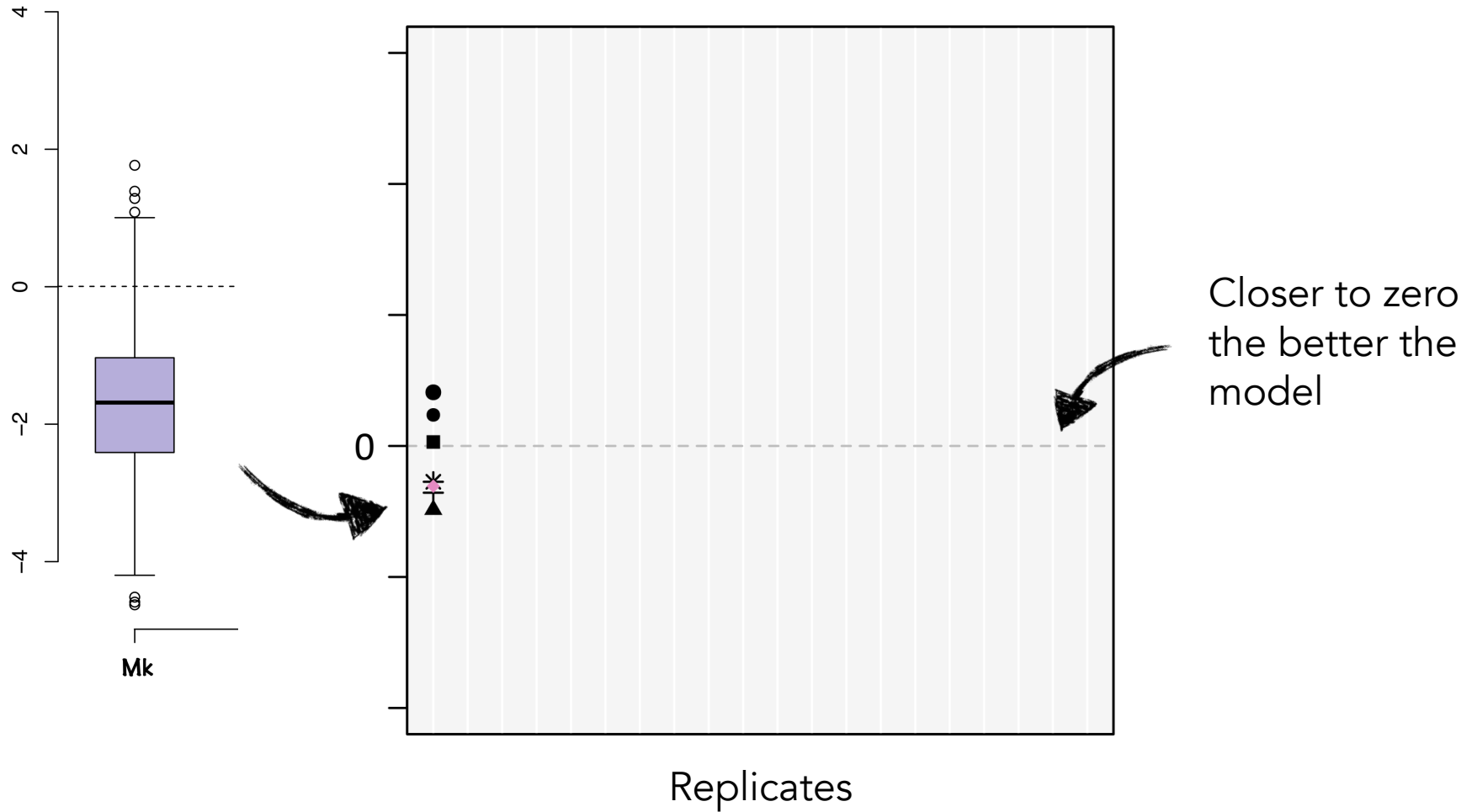
We can use this to
calculate **effect
sizes**

Empirical TS - SimTs

$Sd(\text{All Sim TS})$



Effect Sizes



Data-based test stats

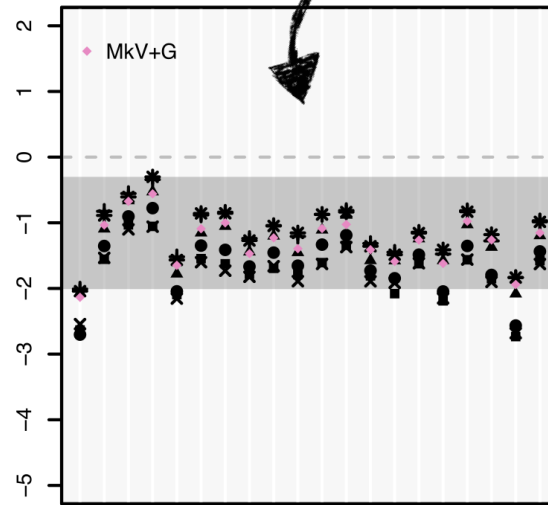


Closer to zero
the better the
model

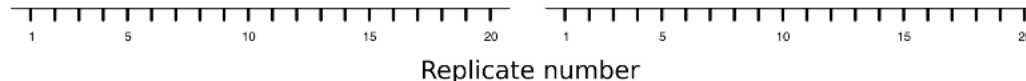
Simulated under
the MkV+G
model:



Simulated under the
MkVP+G model:



We don't see the
correct model
consistently closest
to zero



Grower's Coefficient

Generalised Euclidean
Distances

These test
statistics **are**
not
informative
about the
correct model

Tree-based test stats

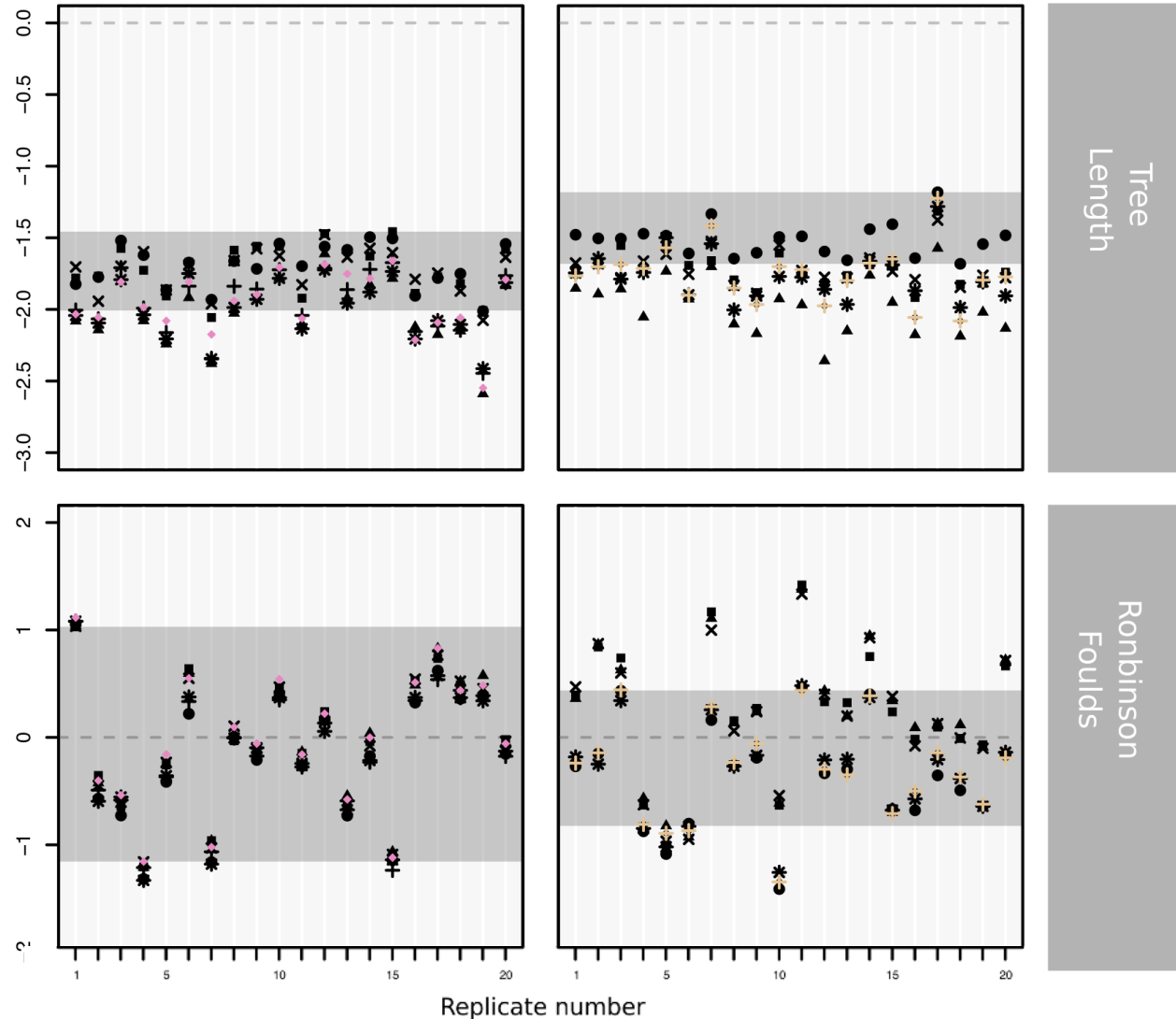
Closer to zero
the better the
model



Simulated under the MkV+G model:



Simulated under the MkVP+G model:



We don't see the
correct model
consistently closest
to zero

These test
statistics **are**
not
informative
about the
correct model

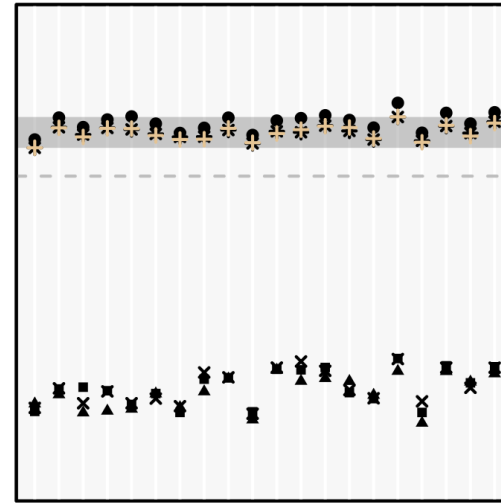
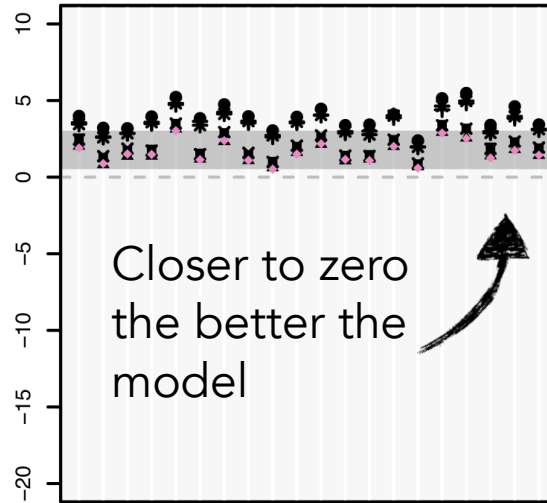
Mixed test stats



Simulated under the MkV+G model:



Simulated under the MkVP+G model:

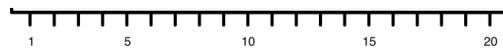


Consistency
Index

Retention
Index

These test statistics **are** informative about the correct model

We do see the correct model consistently closest to zero



Replicate number

Empirical data sets

MkVP+G

MkVP

MkP+G

MkV+G

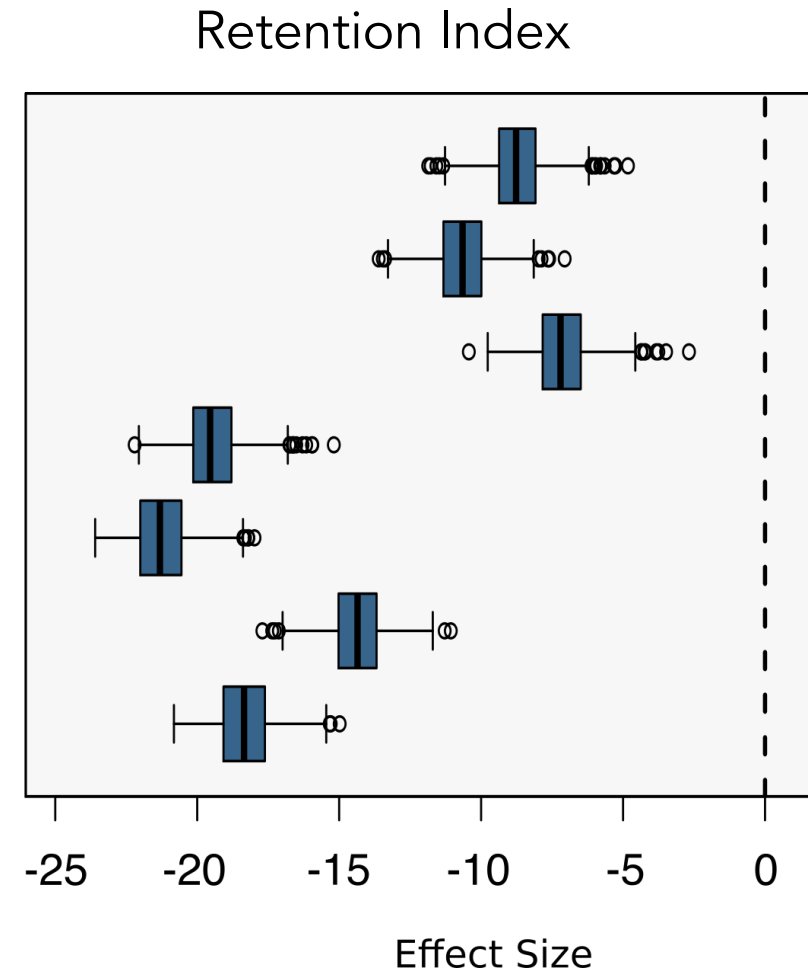
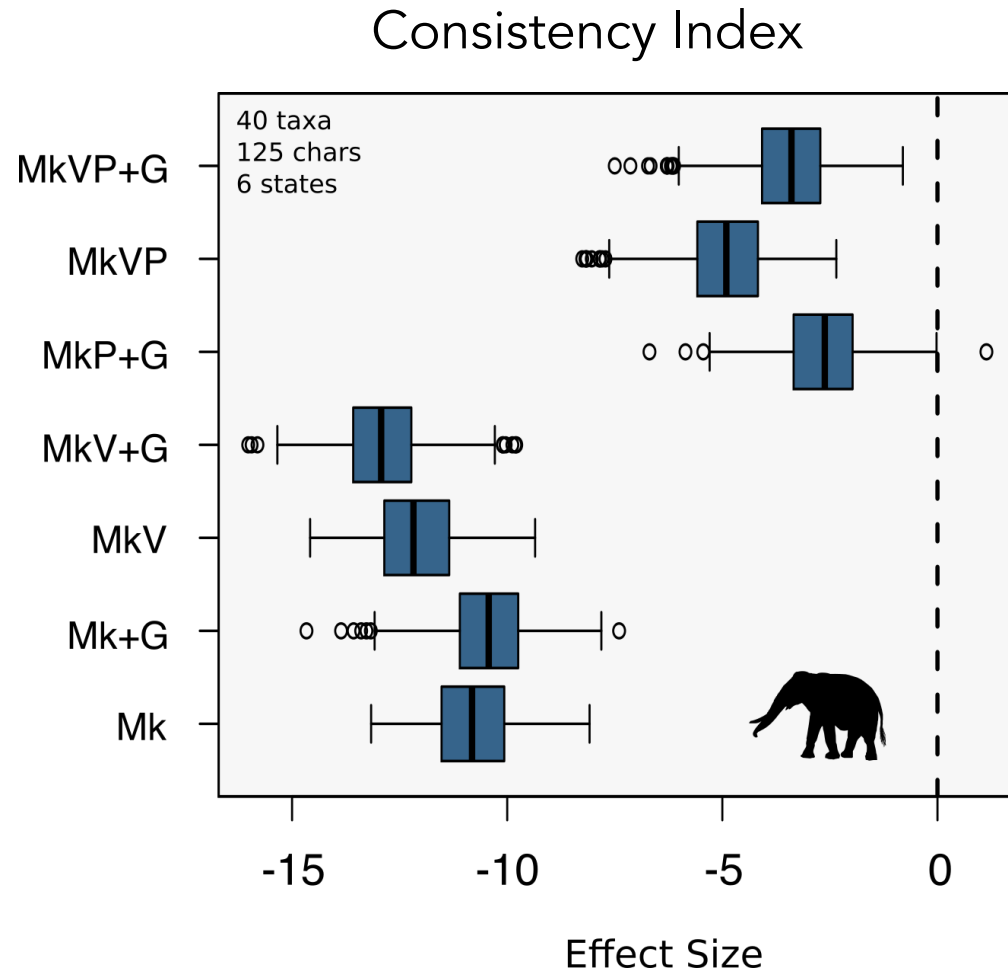
MkV

Mk+G

Mk

found 3
models that
re
dequate

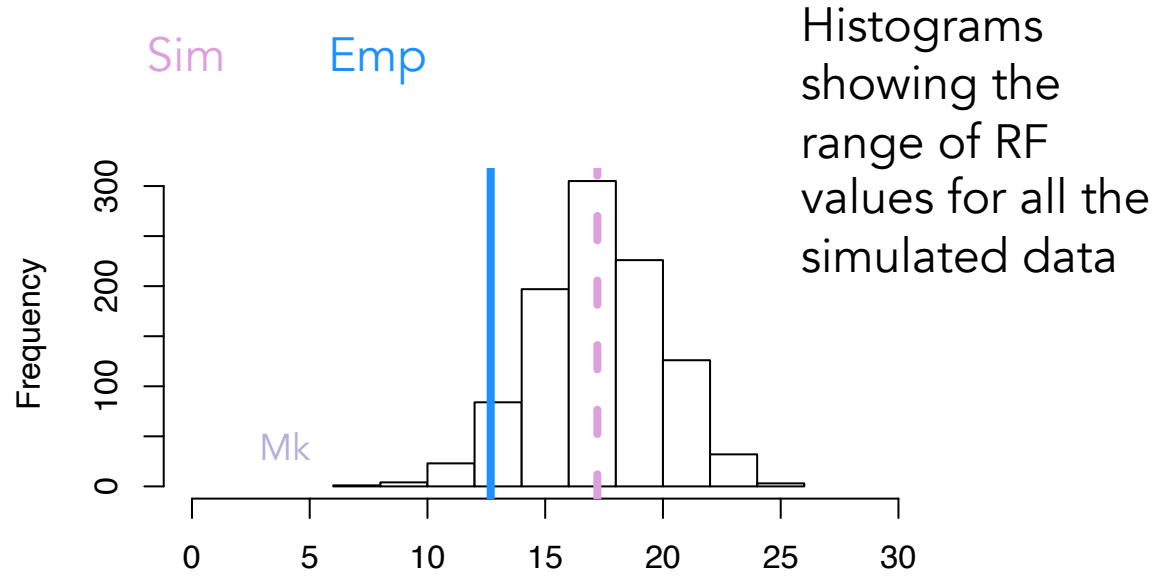
Empirical data sets



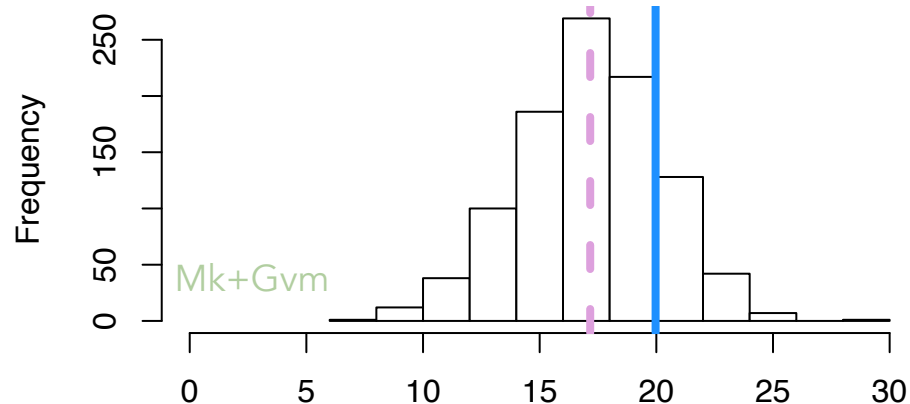
No models
are
adequate

Posterior Predictive P-values

Robinson Foulds Distance



Are these values significantly different from each other?



We will also calculate the P-values in R (look at the midpoint value)

Exercise 2:
Use model adequacy
to determine the
absolute fit of a
model